

Deciphering the Preeclampsia-Specific Immune Microenvironment: The Role of Pro-inflammatory Macrophages

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eLife Assessment

This **valuable** study investigates the immune system's role in pre-eclampsia. The authors map the immune cell landscape of the human placenta and find an increase in macrophages and Th17 cells in patients with pre-eclampsia. Following mouse studies, the authors suggest that the IGF1-IGF1R pathway might play a role in how macrophages influence T cells, potentially driving the pathology of pre-eclampsia. There is **convincing** evidence in this study that will be of interest to immunologists and developmental biologists.

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Abstract

Preeclampsia (PE), a major cause of maternal and perinatal mortality with highly heterogeneous causes and symptoms, is usually complicated by gestational diabetes mellitus (GDM). However, a comprehensive understanding of the immune microenvironment in the placenta of PE and the differences between PE and GDM is still lacking. In this study, Cytometry by time of flight (CyTOF) indicated that the frequencies of memory-like Th17 cells (CD45RA⁺CCR7⁺IL-17A⁺CD4⁺), memory-like CD8⁺ T cells (CD38⁺CXCR3⁺CCR7⁺Helios⁺CD127⁺CD8⁺) and pro-inflam Macs (CD206⁺CD163⁺CD38^{mid}CD107a^{low}CD86^{mid}HLA-DR^{mid}CD14⁺) were increased, while the frequencies of anti-inflam Macs (CD206⁺CD163⁺CD86^{mid}CD33⁺HLA-DR⁺CD14⁺) and granulocyte myeloid-derived suppressor cells (gMDSCs, CD11b⁺CD15^{hi}HLA-DR^{low}) were decreased in the placenta of PE compared with that of NP, but not in that of GDM or GDM&PE. The pro-inflam Macs were positively correlated with memory-like Th17 cells and memory-like CD8⁺ T cells but negatively correlated with gMDSCs. Single-cell RNA sequencing revealed that transferring the F4/80⁺CD206⁺ pro-inflam Macs with a Fcrl2⁺Ccl7⁺Ccl8⁺C1qa⁺C1qb⁺C1qc⁺ phenotype from the uterus of PE mice to normal pregnant mice induced the production of memory-like IL-17a⁺Rora⁺Il1r1⁺TNF⁺Cxcr6⁺S100a4⁺CD44⁺ Th17 cells via IGF1-IGF1R, which contributed to the development and recurrence of PE. Pro-

inflam Macs also induced the production of memory-like CD8⁺ T cells but inhibited the production of Ly6g⁺S100a8⁺S100a9⁺Retnlg⁺Wfdc21⁺ gMDSCs at the maternal-fetal interface, leading to PE-like symptoms in mice. In conclusion, this study revealed the PE-specific immune cell network, which was regulated by pro-inflam Macs, providing new ideas about the pathogenesis of PE.

Introduction

In recent years, low fertility is a global problem (Agbaglo et al., 2022 [\[1\]](#)), exacerbated by pregnancy complications. Preeclampsia (PE) is a progressive systemic disease during pregnancy, with pregnancy-induced hypertension, proteinuria, and liver and kidney injury as the main diagnostic indicators (Chappell et al., 2021 [\[2\]](#)). With a global prevalence of 7–10%, PE is a major cause of maternal and perinatal mortality and morbidity, especially in low-income and middle-income countries. The only specific treatment is delivery (Chappell et al., 2021 [\[2\]](#)). However, due to the variety of symptoms and the heterogeneity of the disease, the pathogenesis of PE is still unclear (Burton et al., 2019 [\[3\]](#); Grotegut, 2016 [\[4\]](#)).

PE is often complicated by gestational diabetes mellitus (GDM), which is characterized by insulin resistance and associated with aberrant maternal immune cell adaption (Corrêa-Silva et al., 2018 [\[5\]](#); McElwain et al., 2021 [\[6\]](#); McIntyre et al., 2019 [\[7\]](#)). Studies have shown that GDM is an independent risk factor for preeclampsia (Nerenberg et al., 2013 [\[8\]](#); Ostlund et al., 2004 [\[9\]](#); Schneider et al., 2012 [\[10\]](#)). However, whether a different pathogenesis underlies between PE and GDM is still unclear.

The fetus is equivalent to a semi-homogenous graft for the mother. Therefore, maternal-fetal immune tolerance plays an important role in the maintenance of pregnancy. Currently, the pivotal mechanism underpinning the pathogenesis of preeclampsia is widely acknowledged to involve an increased frequency of pro-inflammatory M1-like maternal macrophages (Faas et al., 2014 [\[11\]](#); Yao et al., 2019 [\[12\]](#)), along with an elevation in granulocytes that capable of superoxide generation (Lampé et al., 2011 [\[13\]](#); D. Liu et al., 2021 [\[14\]](#)), CD56⁺ CD94⁺ natural killer (NK) cells (Bachmayer et al., 2006 [\[15\]](#); Travis et al., 2020 [\[16\]](#)), CD19⁺CD5⁺ B1 lymphocytes (Jensen et al., 2012 [\[17\]](#); LaMarca et al., 2011 [\[18\]](#); Zhong et al., 2007 [\[19\]](#)), and activated γδ T cells (Chatterjee et al., 2017 [\[20\]](#)). Conversely, this pathological process is accompanied by a notable decrease in the frequency of anti-inflammatory M2-like macrophages (Faas et al., 2014 [\[11\]](#); Yao et al., 2019 [\[12\]](#)) and NKp46⁺ NK cells (Fukui et al., 2011 [\[21\]](#)). Adaptive immune response is also critical for the pathogenesis of PE (Deer et al., 2023 [\[22\]](#)). In PE, CD4⁺ T cells prefer to differentiate into pro-inflammatory Th1 and Th17 (Eghbal-Fard et al., 2019 [\[23\]](#); Fu et al., 2014 [\[24\]](#); Lang et al., 2021 [\[25\]](#); Lu et al., 2020 [\[26\]](#); Saito et al., 2007 [\[27\]](#)) instead of non-inflammatory Th2 and Treg (Care et al., 2018 [\[28\]](#); Cornelius et al., 2015a [\[29\]](#); Saito et al., 2007 [\[27\]](#); Santner-Nanan et al., 2009 [\[30\]](#); Sasaki et al., 2007 [\[31\]](#)).

Moreover, PE can be subdivided into early- and late-onset PE diagnosed before 34 weeks or from 34 weeks of gestation, respectively. Research has revealed that among the myriad of cellular alterations in PE, pro-inflammatory M1-like macrophages and intrauterine B1 cells display an augmented presence at the maternal-fetal interface of both early-onset and late-onset PE patients. Decidual natural killer (dNK) cells and neutrophils emerge as paramount contributors, playing a more crucial role in the pathogenesis of early-onset PE (Aneman I et al., 2020 [\[32\]](#)). However, a comprehensive and in-depth understanding of the maternal-fetal interface of PE is still lacking.

The immune cells crosstalk with each other at the maternal-fetal interface elaborately. For example, the interaction between B2 cells and Th cells depends on CD40 and its ligand CD40L, which helps B2 cells provide AT1-AAAs and differentiate into memory B cells in reducing uterine perfusion pressure (RUPP) rats (Cornelius et al., 2015b [\[33\]](#)). The cytokines derived from NK cells can promote the differentiation of Th17 to co-work in recurrent miscarriage (Fu et al., 2013 [\[34\]](#)). The

intricate interplay and precise coordination between immune cells are pivotal in maintaining pregnancy. Nevertheless, the regulation of the immune network at the maternal-fetal interface of PE patients still needs to be further studied.

In this study, we combined cytometry by time of flight (CyTOF), single-cell RNA sequencing (scRNA-seq), and rodent experiment to identify the overall immune cell composition and their interactions at the maternal-fetal interface in PE, GDM, and GDM complicated with PE (GDM&PE). This study revealed the PE-specific immune cell network, which was regulated by pro-inflammatory macrophages, providing new ideas about the pathogenesis of PE.

Results

Overall immune cell profile in the placenta of PE, GDM, and GDM&PE

To fully characterize the immune microenvironment at the maternal-fetal interface of PE, we collected full-term placentas of PE and normal pregnancy (NP), and did a CyTOF test ([Table 1](#)), which panel was shown in [Table 2](#). Since PE is usually complicated by GDM, the subjects of GDM and GDM complicated with PE (GDM&PE) were also included. The experimental workflow for CyTOF is shown in the schematic diagram ([Figure 1A](#)). We found that in addition to the well-known macrophages and T cells, there are also $\gamma\delta$ T cells, B cells, NK cells, granulocytes, dendritic cells (DCs), and myeloid-derived suppressor cells (MDSCs). An overall distribution of CD45⁺ cell subsets in placenta as shown in the t-SNE maps ([Figure 1B](#)). There were no significant differences in the proportion of these large subpopulations comparing the PE, GDM, GDM&PE group to the NP group ([Figure 1C](#)). To identify the ten cell subpopulations, CD4, CD8, and $\gamma\delta$ TCR were used as markers to distinguish CD4⁺ T cells, CD8⁺ T cells, and $\gamma\delta$ T cells. CD11b is a marker that distinguishes myeloid cells, including macrophages and granulocytes ([Figure 1D](#)). The ten cell subsets were further annotated by 41 markers in the heatmap ([Figure 1E](#), [Table 3](#)).

Specific altered T cell profile in the placenta of PE

To gain a comprehensive understanding of the distribution patterns within each cellular subset, we conducted separate clustering analyses for CD4⁺T, CD8⁺T, and $\gamma\delta$ T cells, respectively. We first analyzed 15 clusters of CD4⁺ T cells from the placentas of NP, PE, GDM, and GDM&PE ([Figure 2A](#)). CD4⁺ T cell clusters were defined by canonical marker set signatures ([Figure 2-figure supplement 1A](#)). The frequencies of memory-like CD45RO⁺CCR7⁺CD69⁺CD127^{low}TIGIT⁺Helios⁺ CD4⁺ T cells (cluster 8) were significantly increased in individuals with PE compared with NP controls. At the same time, there was no significance in the GDM and GDM&PE groups compared with the NP group ([Figure 2B](#), [Table 4](#)). We analyzed the expression of common intracellular molecules in CD4⁺ memory-like T cells by sorting CD45RO⁺CCR7⁺CD4⁺ T cells from placental samples by flow cytometry. Higher levels of IL-17A and lower levels of Foxp3 were found in memory-like CD45RO⁺CCR7⁺CD4⁺ T cells in the placentas of individuals with PE ([Figure 2C](#)). Consistent with the results of CyTOF, higher fluorescence intensity of CD45RO⁺CD4⁺ T cells was found in the placentas of individuals with PE than that of NP, and this cell population was mainly located in the placental sinusoids ([Figure 2D](#)), suggesting that these T cells should be recently recruited. Moreover, lower expression of immune checkpoint molecules including T-cell immunoglobulin mucin-3 (Tim-3) and programmed cell death 1 (PD-1) was found in CD45RO⁺CCR7⁺CD4⁺ memory T cells in the PE group, suggesting that these cells have a lower immunosuppressive capacity ([Chen et al., 2022](#); [Fanelli et al., 2021](#); [Rasmussen et al., 2022](#); [Wang et al., 2016](#)), there was no significant difference in the GDM or GDM&PE groups comparing with the NP group ([Figure 2E](#)). Then, 18 clusters of CD8⁺ T cells were analyzed ([Figure 2F](#)). CD8⁺ T cell clusters annotation was based on the expression of canonical marker

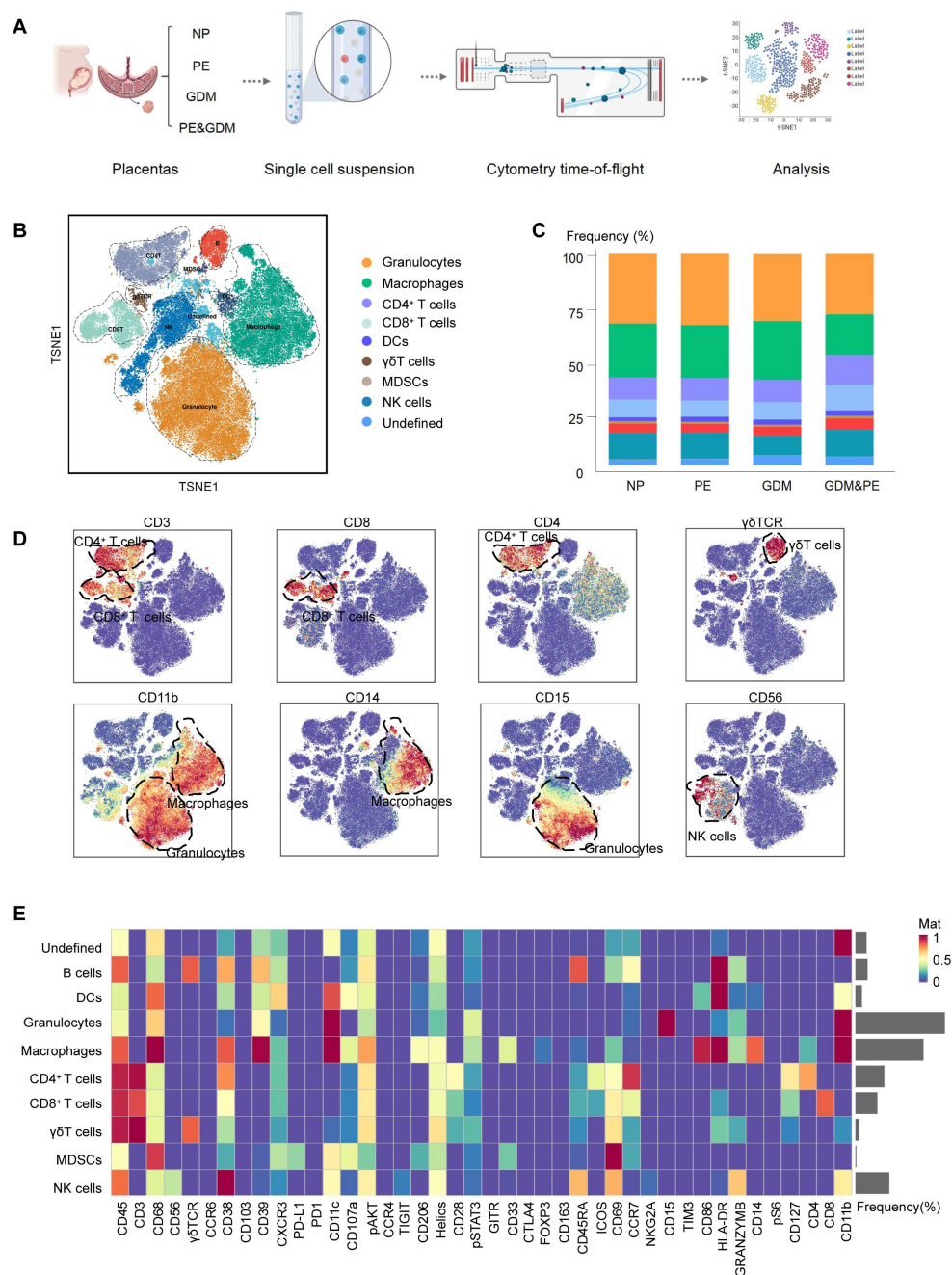


Figure 1

Identification and characterization of placental immune cells using CyTOF

(A) Schematic of the experimental workflow in CyTOF experiment. The placentas were obtained from individuals with normal pregnancy (NP, n=9), preeclampsia (PE, n=8), gestational diabetes mellitus (GDM, n=8) or GDM&PE (n=7).

(B) t-SNE maps showing 8×10^4 CD45⁺ cells (the average cell number of the all samples) from the placenta overlaid with color-coded clusters and the distributions of B cells, CD4⁺ T cells, CD8⁺ T cells, DC, $\gamma\delta$ T cells, monocytes, granulocytes, MDSC, and NK cells.

(C) Percentages of each cell type of CD45⁺ cells in placentas.

(D) t-SNE maps showing the expression of CD3, CD8, CD4, $\gamma\delta$ TCR, CD14, CD15, CD56.

(E) Heatmap showing the expression levels of markers in CD45⁺ cell subsets.

Data were compared between NP and PE, NP and GDM, NP and GDM&PE using the Kruskal-wallis test and represented as mean $\pm$ SEM (*P < 0.05, ** P < 0.01, *** P < 0.001; NS, not significant).

Parameters	NP (n=9)	PE (n=8)	GDM (n=8)	GDM&PE (n=7)	p Value
Age (years)	31.33 ± 2.108	31.75 ± 4.206	35.25 ± 3.419	32.43 ± 2.321	0.1008
BMI (kg/m ²)	27.35 ± 1.740	30.84 ± 2.888	29.56 ± 2.689	30.16 ± 2.280	0.0518
Gestational age (weeks)	38.89 ± 0.558	37.39 ± 1.253	38.73 ± 0.419	36.22 ± 1.372	<0.001
Number of living children	0.4444 ± 0.497	0.500 ± 0.500	1.125 ± 0.331	0.429 ± 0.495	0.0236
Previous abortions	0.7778 ± 1.030	1.000 ± 0.866	1.000 ± 1.000	1.714 ± 2.050	0.5848
Mean Systolic Blood Pressure	109.7 ± 5.400	151 ± 7.225	106.9 ± 2.848	150.7 ± 9.161	<0.0001

Table 1

Details of the individuals included in the CyTOF

Table 2

CytoF antibody panel used for analyzing placentas from individuals with normal pregnancy (NP), preeclampsia (PE), gestational diabetes mellitus (GDM) and GDM complicated with PE (GDM&PE)

Target	Metal tag
CD45	HI30
CD3	UCHT1
CD68	Y1/82A
CD56	NCAM16.2
gd TCR	5A6.E9
CD19	HIB19
CCR6	G034E3
CD38	HIT2
CD103	B-Ly7
CD39	A1
CXCR3	G025H7
PD-L1	29E.2A3
PD-1	EH12.2H7
CD11C	BU15
CD107a	H4A3
pAKT	D9E
CCR4	L291H4
TIGIT	A15153G
CD206	15-2
Helios	22F6
CD28	CD28.2
pSTAT3	4/p-Stat3
GITR	110416
CD33	WM53
CTLA-4	BN13
FOXP3	PCH101
CD163	GHI/61
CD45RA	HI100
ICOS	C398.4A
CD69	FN50
CCR7	G043H7
NKG2A	131411
CD15	W6D3
TIM-3	F38-2E2
CD86	Fun-1
HLA-DR	L243
Granzyme B	QA16A02
CD14	M5E2
pS6	A17020B
CD127	A019D5
CD4	RPA-T4
CD8	RPA-T8
CD11b	M1/70

Table 3

Expression levels of markers identified in each immune subset in the placentas

	B cells	CD4 ⁺ T	CD8 ⁺ T	gdTCR	NK cells	Macrophage	Granulocyte	MDSC	Dendritic cells
CD45	++	++	++	++	++	++	++	++	++
CD3	-	++	++	-	-	-	-	-	-
CD68	+	+	+	+	+	++	+	++	++
CD56	-	-	-	-	+	-	-	-	-
gd TCR	++	-	-	++	-	-	-	-	-
CD19	++	-	-	++	-	-	-	-	-
CCR6	+	-	-	-	-	-	-	-	-
CD38	++	++	+	+	++	++	-	-	+
CD103	-	-	-	-	-	-	-	-	-
CD39	++	-	-	-	+	++	+	+	+
CXCR3	+	+	+	+	+	+	-	+	++
PD-L1	-	-	-	-	-	-	-	+	-
PD-1	-	-	-	-	-	-	-	-	-
CD11C	+	-	-	-	+	++	++	+	++
CD107a	+	-	-	-	-	+	-	+	+
pAKT	+	+	+	+	+	++	+	+	+
CCR4	-	-	-	-	-	-	-	-	-
TIGIT	-	-	-	-	-	-	-	-	-
CD206	-	-	-	-	-	+	-	+	-
Helios	+	+	+	+	+	+	+	+	+
CD28	-	+	+	+	-	-	-	-	-
pSTAT3	+	+	-	+	-	+	+	-	-
GITR	-	-	-	-	-	-	-	-	-
CD33	-	-	-	-	-	+	-	+	-
CTLA-4	-	-	-	-	-	-	-	-	-
FOXP3	-	-	-	-	-	-	-	-	-
CD163	-	-	-	-	-	-	-	-	-
CD45RA	++	+	+	+	++	-	-	-	-
ICOS	-	+	+	-	-	-	-	-	-
CD69	+	+	+	+	++	+	-	++	-
CCR7	+	++	+	+	-	-	-	+	-
NKG2A	-	-	-	-	-	-	-	-	-
CD15	-	-	-	-	-	-	++	-	-
TIM-3	-	-	-	-	-	-	-	-	-
CD86	-	-	-	-	-	++	-	-	+
HLA-DR	++	-	+	+	-	++	+	-	++
Granzyme	+	-	-	+	++	+	+	-	-
CD14	-	-	-	-	-	++	-	-	-
pS6	-	-	-	-	-	-	-	-	-
CD127	-	++	+	-	-	-	-	-	-
CD4	-	++	-	-	-	-	-	-	-
CD8	-	-	++	-	-	-	-	-	-
CD11b	-	-	-	+	+	++	++	-	+

signatures (**Figure 2-figure supplement 1B**). CD38⁺CXCR3[−]CCR7⁺Helios[−]CD127[−] (cluster 2) memory-like CD8⁺ T cells were significantly increased in the PE group (**Figure 2G**, **Table 4**). CD38⁺CXCR3⁺CCR7⁺Helios[−]CD127[−] memory-like CD8⁺ T cells of the PE group expressed lower levels of PD-1 and TIGIT, suggesting the activation of cytotoxicity of these cells (**Figure 2H**) (Morita et al., 2020). Then, 22 clusters of $\gamma\delta$ T cells were analyzed and annotated (**Figure 2-figure supplement 1C-D**). Cluster 15 (Helios^{mid}CD28^{mid}CD69^{mid}HLA-DR^{mid}CD127^{mid} $\gamma\delta$ T cells), with lower expression of Tim3, was significantly decreased in the PE group (**Figure 2-figure supplement 1E-F**). Nevertheless, neither the CD4⁺ T cells nor the CD8⁺ T cells exhibited any alteration in the memory-like clusters when comparing the GDM or GDM&PE groups to the NP group.

In conclusion, significant changes in placental T cell profile were found in the placentas of PE but not in those of GDM or GDM&PE, suggesting that abnormal activation of T cells in the placenta is associated with the pathogenesis of PE.

Abnormal polarization of macrophages was correlated with specific immune cell subsets in individuals with PE

Except for T cells, we also analyzed 29 clusters of CD45⁺CD3[−]CD11b⁺ cells from placentas of NP, PE, GDM, and GDM&PE, including 9 clusters of macrophages, 11 clusters of granulocytes, and 5 clusters of NK/NK-like cells (**Figure 3A**). The clusters of CD45⁺CD3[−]CD11b⁺ cells were defined by canonical marker set signatures (**Figure 3B**). Moreover, significantly decreased frequencies of CD68^{mid}CD39^{mid}HLA-DR^{low}CD11b⁺CD15⁺ granulocytes (cluster 12), which are identified as granulocyte myeloid-derived suppressor cells (gMDSCs), were found in the PE group (**Figure 3C**, **Table 4**). In addition, the frequency of anti-inflammatory macrophages (anti-inflam Macs) (CD206⁺CD163[−]CD86^{mid}CD33⁺HLA-DR⁺, cluster 25) was also significantly decreased, whereas the frequency of pro-inflammatory macrophages (pro-inflam Macs) (CD206[−]CD163[−]CD38^{mid}CD107a^{low}CD86^{mid}HLA-DR^{mid}CD14⁺, cluster 23) was significantly increased in the PE group (**Figure 3C**, **Table 4**). However, the frequencies of macrophages and gMDSCs were unchanged significantly in the GDM or GDM&PE groups.

Pearson correlation analysis indicated positive correlations between pro-inflam Macs (cluster 23 in CD11b⁺ cells) and CD4⁺ memory-like T cells (cluster 8 of CD4⁺ T cells), as well as CD8⁺ memory-like T cells (cluster 2 in CD8⁺ T cells). CD206[−] pro-inflam Macs were negatively correlated with gMDSCs (cluster 12 in CD11b⁺ cells), but were not statistically significant. Conversely, CD206⁺ anti-inflam Macs (cluster 25 in CD11b⁺ cells) were positively correlated with gMDSCs and negatively correlated with CD8⁺ memory-like T cells (**Figure 3D, 3E**).

These results suggested that abnormally polarized CD206[−] pro-inflam Macs are positively correlated with memory-like CD4⁺ and CD8⁺ T cells in the placentas of individuals with PE and negatively correlated with gMDSCs.

F4/80⁺CD206[−] pro-inflam Macs induced immune imbalance at the maternal-fetal interface and PE-like symptoms

Though increased pro-inflam Macs have been reported in the placenta from individuals with PE (Faas et al., 2014), few studies have reported the interaction between macrophages and other immune cells in the placenta. In this study, transcriptome RNA-seq was used to analyze the difference between CD45⁺F4/80⁺CD206[−] pro-inflam Macs and CD45⁺F4/80⁺CD206⁺ anti-inflam Mac isolated from the uterus and placentas of mice with reducing uterine perfusion pressure (RUPP) (**Figure 4A**). The diagrammatic representation of the RUPP model is presented in **Figure 4-figure supplement 1A**. To establish the PE mouse model, we ligated uterine arteries in pregnant mice on day 12.5 of gestation, and mice with sham operation were considered as controls (**Figure 4-figure supplement 1A**). We found that mice of RUPP showed an increased embryo absorption

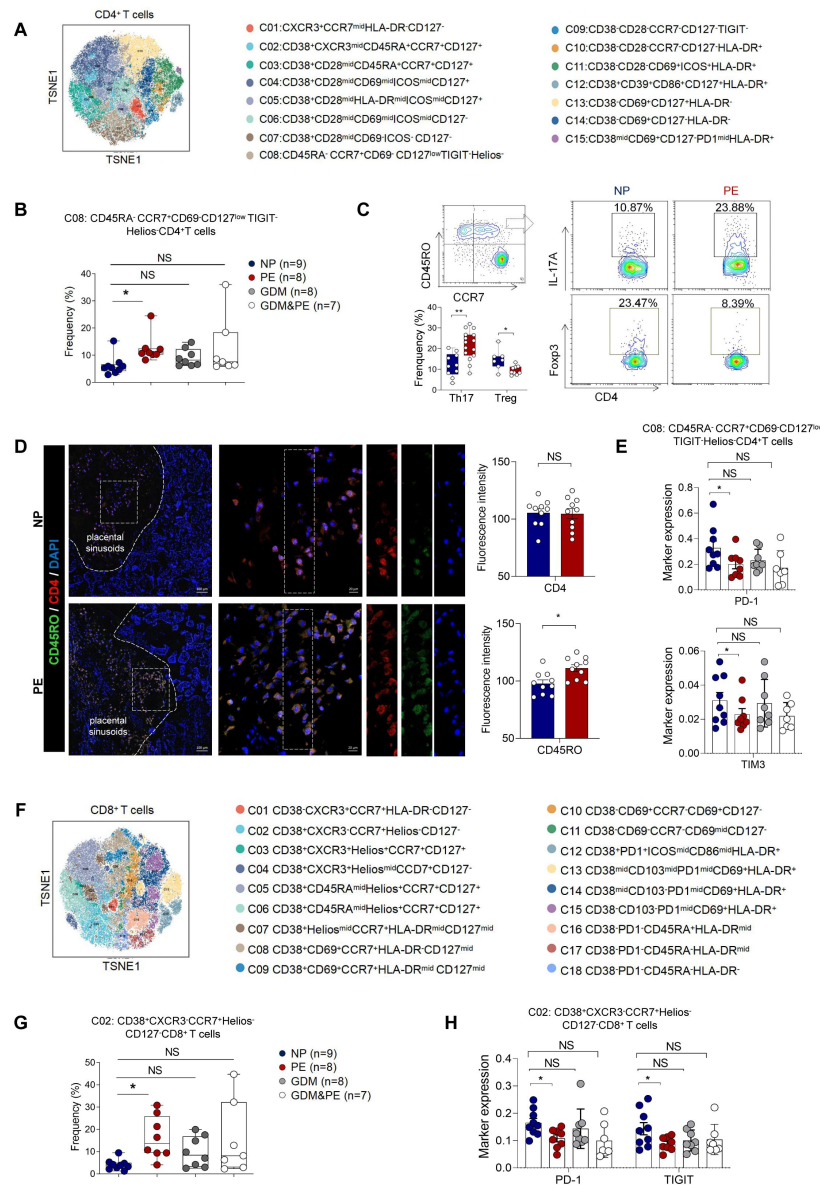


Figure 2

Specific altered T cell profile in the placentas of individuals with PE

(A) Distribution of the CD4⁺ T cells in the group analyzed using t-SNE.

(B) Scatter dot plots showing the frequencies of cluster 8 of CD4⁺ T cells in the placentas of individuals with NP, PE, GDM and GDM&PE (n=9 in NP group; n=8 in PE group; n=8 in GDM group; n=7 in GDM&PE group).

(C) Expression of IL-17A and Foxp3 in CD45RO⁺CCR7⁺CD4⁺ T cells in placentas of individuals with NP and PE using flow cytometry (IL-17A: n=10 in NP group, n=15 in PE group; Foxp3: n=7 in NP group, n=9 in PE group).

(D) Immunofluorescence co-staining of CD4 (red), CD45RO (green) and DAPI (blue) in frozen placental sections. The right panels show the fluorescence intensity of CD4 and CD45RO. Scale bar, 20 μ m.

(E) Scatter dot plots showing significantly altered markers of in cluster 8 of CD4⁺ T cells.

(F) Distribution of the CD8⁺ T cells analyzed using t-SNE.

(G) Scatter dot plots showing the frequencies of cluster 2 of CD8⁺ T cells in the placentas of individuals with NP, PE, GDM and GDM&PE.

(H) Scatter dot plots showing significantly altered markers in cluster 2 of CD8⁺ T cells.

Data were compared between NP and PE, NP and GDM, NP and GDM&PE using the Kruskal-wallis test and represented as mean $\pm$ SEM (*P < 0.05, ** P < 0.01, *** P < 0.001; NS, not significant).

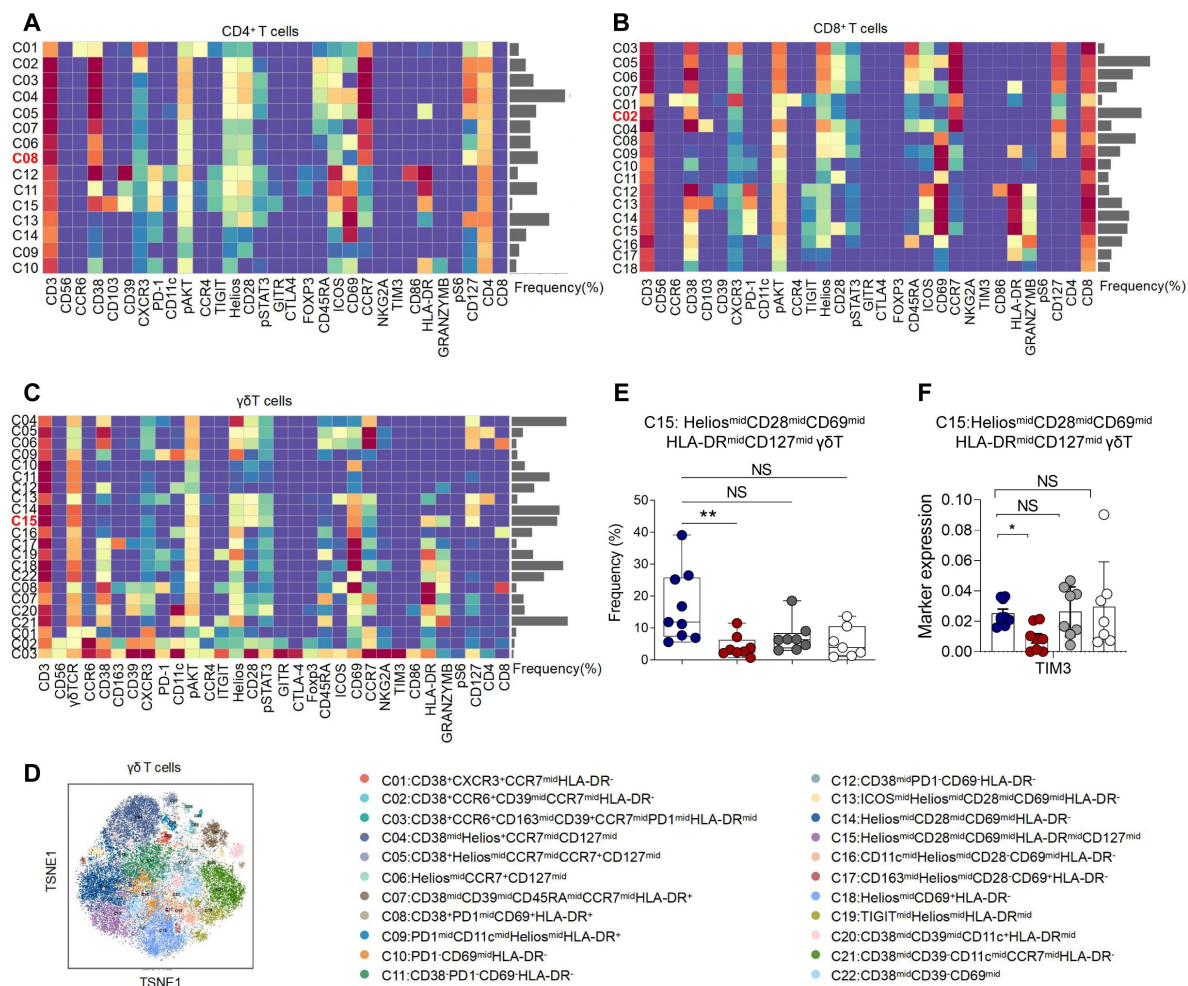


Figure 2—figure supplement 1

Identification of the placental T cell subsets.

(A) Heatmap showing the expression levels of markers in the CD4⁺ T subsets (n=9 in NP group; n=8 in PE group; n=8 in GDM group; n=7 in GDM&PE group).

(B) Heatmap showing the expression levels of markers in the CD8⁺ T subsets.

(C) Heatmap showing the expression levels of markers in the γδT subsets.

(D) Distribution of the γδT cells in the group analyzed using t-SNE.

(E) Scatter dot plots showing the frequencies of cluster 15 of γδT cells in the placentas of individuals with NP, PE, GDM and GDM&PE.

(F) Scatter dot plots showing significantly altered markers in cluster 15 of γδT cells.

Data were compared between NP and PE, NP and GDM, NP and GDM&PE using the Kruskal-wallis test and represented as mean±SEM (*P < 0.05, ** P < 0.01, *** P < 0.001; NS, not significant).

Table 4

The marker profile of PE-specific immune subsets

	memory-like CD4 ⁺ T cells	memory-like CD8 ⁺ T cells	gMDSCs	pro-inflammatory macrophages	anti-inflammatory macrophages
	Cluster 08 of CD4 ⁺ T cells	Cluster 02 of CD8 ⁺ T cells	Cluster 12 of CD3 ⁺ CD11b ⁺ cells	Cluster 23 of CD3 ⁺ CD11b ⁺ cells	Cluster 25 of CD3 ⁺ CD11b ⁺ cells
CD45	++	++	++	++	++
CD3	++	++	-	-	-
CD68	-	-	+	++	++
CD56	-	-	-	-	-
gd TCR	-	-	-	-	-
CD19	-	-	-	-	-
CCR6	-	-	-	-	-
CD38	++	++	-	+	+
CD103	-	-	-	+	+
CD39	+	-	+	-	-
CXCR3	+	+	-	-	-
PD-L1	-	-	-	-	-
PD-1	-	-	-	-	-
CD11C	-	-	++	++	++
CD107a	-	-	-	+	+
pAKT	+	+	-	-	-
CCR4	-	-	-	-	-
TIGIT	-	-	-	-	-
CD206	-	-	+	-	+
Helios	+	+	-	-	-
CD28	+	+	-	-	-
pSTAT3	-	-	-	-	-
GITR	-	-	-	-	-
CD33	-	-	-	-	+
CTLA-4	-	-	-	-	-
FOXP3	-	-	-	-	-
CD163	-	-	-	-	-
CD45RA	-	-	-	-	-
ICOS	+	+	-	-	-
CD69	+	-	-	-	-
CCR7	++	++	-	-	-
NKG2A	-	-	-	-	-
CD15	-	-	++	-	-
TIM-3	-	-	-	-	-
CD86	-	-	-	+	+
HLA-DR	-	-	+	+	++
Granzyme B	-	-	+	-	-
CD14	-	-	-	+	+
pS6	-	-	-	-	-
CD127	+	-	-	-	-
CD4	++	-	-	-	-
CD8	-	++	-	-	-
CD11b	-	-	++	++	++

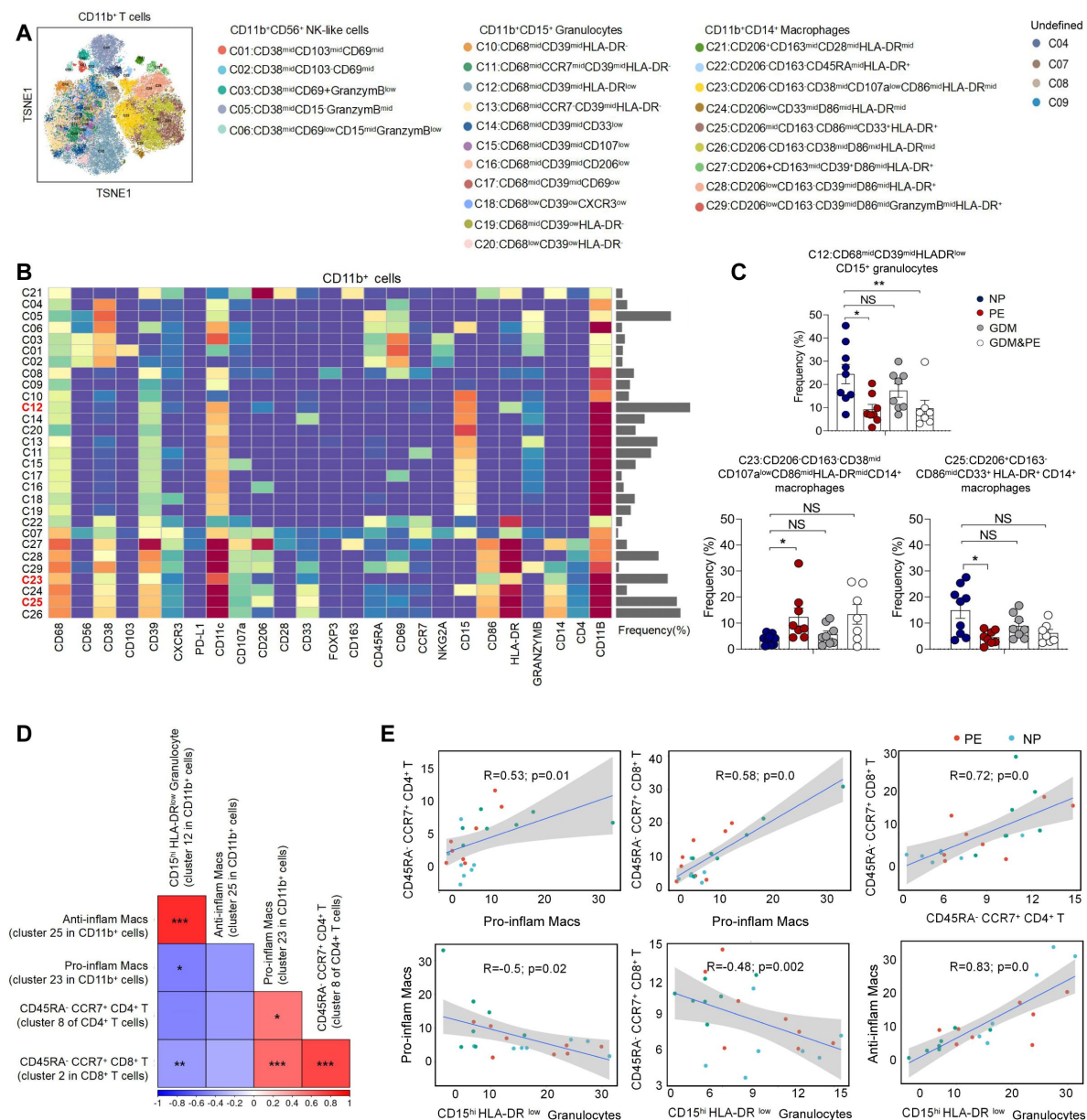


Figure 3

Identification of the placental CD11b⁺ cell subsets and the Interaction between placental immune cells.

(A) Distribution of the CD11b⁺ immune cells in group analyzed using t-SNE.

(B) Heatmap showing the expression levels of markers in the CD11b⁺ cells.

(C) Scatter dot plots showing the frequencies of cluster 12, cluster 23, and cluster 25 of CD11b⁺ cells in the placentas.

(D) Interaction between placental immune cells showed in heatmap.

(E) Scatter plots of Pearson's correlation analysis between placental immune cells.

Data were compared between NP and PE, NP and GDM, NP and GDM&PE using the Kruskal-wallis test and represented as mean±SEM (*P < 0.05, ** P < 0.01, *** P < 0.001; NS, not significant).

rate, decreased fetal weight, and pup crown-rump length compared with the sham operation group (**Figure 4-figure supplement 1B**). Increased systolic blood pressure (SBP) and urine albumin creatine ratio (UACR) were also observed in mice with RUPP, which indicated a successful PE mice mode was built (**Figure 4-figure supplement 1C**). Then, we injected the PE-mice derived macrophages into normal pregnant mice. An increased embryo resorption rate, decreased fetal top-rump length and fetal weight were found in mice injected with PE-mice derived macrophages compared with mice injected PBS (**Figure 4-figure supplement 1D**). Increased SBP and UACR were also found in mice injected with PE-mice derived macrophages (**Figure 4-figure supplement 1E**).

In addition, the green fluorescent protein (GFP) pregnant mice were used to distinguish between maternal and fetal derived macrophages. The wild-type (WT) female mice were mated with either transgenic male mice, genetically modified to express GFP to generate either GFP-expressing pups (GFP⁺pups), or mated with WT male mice for control (**Figure 4-figure supplement 1F**). We found that the majority of macrophages in the uterus and placenta are of maternal origin (CD11b⁺GFP⁺). In contrast, fetal-derived macrophages (CD11b⁺GFP⁺) represent a mere fraction of the total macrophage population (**Figure 4-figure supplement 1G**). In light of these findings, we have incorporated mouse uterine tissues to isolate macrophages for subsequent experiments with enhanced effectiveness.

Significant differences were found between the pro-inflam and anti-inflam macrophages derived from RUPP mouse (**Figure 4B**). There are 2123 genes increased, and 2699 genes decreased significantly in the pro-inflam Macs compared to the anti-inflam Macs (**Figure 4C**).

The expression levels of genes associated with inflammatory response (Ccl7, Ccl8, Ccl2, IL6) (He et al., 2019; Wu et al., 2023), complement system activation (C1qa, C1qb) (Chen et al., 2021), and lipid metabolism (Pltp, Apoe) (Desrumaux et al., 2016) were significantly increased in the pro-inflam Macs; whereas the expression levels of genes associated with the regulation of angiogenesis and vascular endothelial growth factor production (VEGFα, Macro) and tissue development (TGF-β1, Thbs1, Fn1, Slpi) (Jin et al., 2023; Nugteren et al., 2021; Li et al., 2022) were significantly decreased (**Figure 4C, 4D**). Moreover, Gene Set Enrichment Analysis (GSEA) was performed to explore the most significantly enriched functional terms between the two groups of macrophages. We found that NF-κB signaling and interspecies interaction between organisms were enriched in the pro-inflam Macs, while tissue development and epithelial cell morphogenesis were enriched in the anti-inflam Macs (**Figure 4E**).

To further investigate the effect of the pro-inflam Macs in immune imbalance at the maternal-fetal interface, PLX3397, the inhibitor of CSF1R, which is needed for macrophage development, was used to deplete the macrophages of pregnant mice (X. Chen et al., 2023; J. Chen et al., 2022) (**Figure 4-figure supplement 2A**). As expected, an increased embryo resorption rate, decreased fetal top-rump length and fetal weight were found in mice injected with pro-inflam Macs (**Figure 4F**). Increased SBP and UACR were also found in mice injected with pro-inflam Macs (**Figure 4G**). Moreover, it is shown that CD44⁺ memory-like Th17 cells and memory-like CD8⁺ T cells increased while CD11b⁺Ly6G⁺ gMDSCs decreased in mice injected pro-inflam Macs compared with mice injected anti-inflam Macs, which validated our findings with CyTOF (**Figure 4H**). Clodronate liposomes were also used to deplete the macrophages of pregnant mice (X. Liu et al., 2022) before pro-inflam Macs or anti-inflam Macs were injected into the mice. The same experimental results were obtained (**Figure 4-figure supplement 2B-2E**). In conclusion, the F4/80⁺CD206⁻ pro-inflam Macs induced immune imbalance at the maternal-fetal interface and PE-like symptoms.

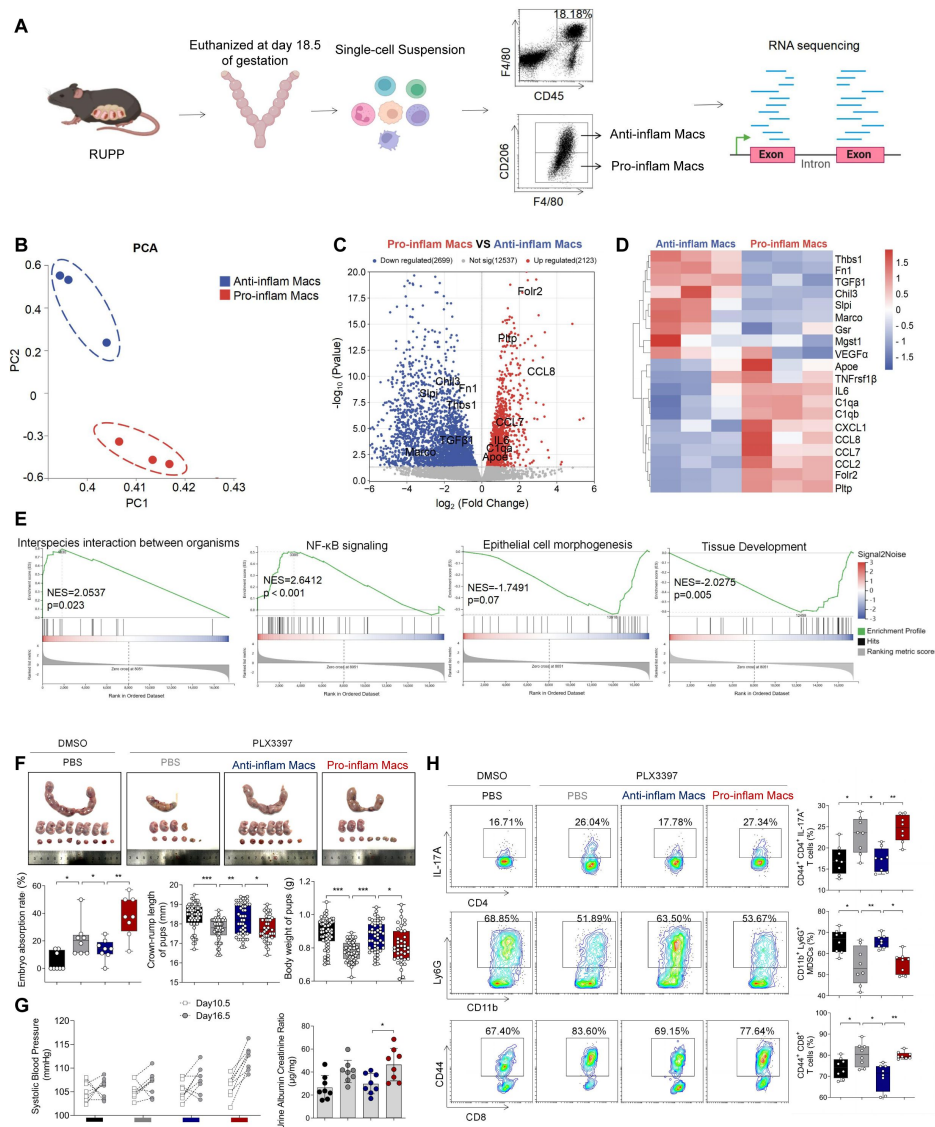


Figure 4

The immune imbalance at the maternal-fetal interface induced by F4/80⁺CD206⁻ pro-inflam Macs.

(A) An illustration showcasing the RNA-seq of the CD45⁺F4/80⁺CD206⁻ pro-inflam Macs and CD45⁺F4/80⁺CD206⁺ anti-inflam Macs derived from the RUPP mouse model.

(B) Principal component analysis (PCA) reflected the differences between the two groups of macrophages (n=3).

(C) The volcano map shows a comparison of the content and P value of gene expression between pro-inflam Macs and anti-inflam Macs. Differential expression genes were screened out when P < 0.05. Red dots indicate genes with increased expression in pro-inflam Macs. Blue dots indicate genes with decreased expression.

(D) The volcano map shows differential expression genes between pro-inflam Macs and anti-inflam Macs.

(E) Representative pathways enriched in the identified genes as determined by GSEA (p value < 0.05).

(F) Embryo abortion rate of the pregnant mice, body weight and crown-rump length of pups measured on day 18.5 of gestation. Black represents mice treated with DMSO (n=8); gray represents mice treated with PLX3397 (n=8); blue represents mice injected with CD45⁺F4/80⁺CD206⁺ anti-inflammatory macrophages (n=8); red represents mice injected with CD45⁺F4/80⁺CD206⁻ pro-inflammatory macrophages (n=8).

(G) SBP and UACR of pregnant mice in the four groups.

(H) Frequencies of CD44⁺CD4⁺IL-17A⁺ cells, CD44⁺CD8⁺ T cells and CD11b⁺Ly6G⁺ granulocytes analyzed by flow cytometry. Data were compared between groups using the Kruskal-wallis test and represented as mean ± SEM (*P < 0.05, **P < 0.01, ***P < 0.001).

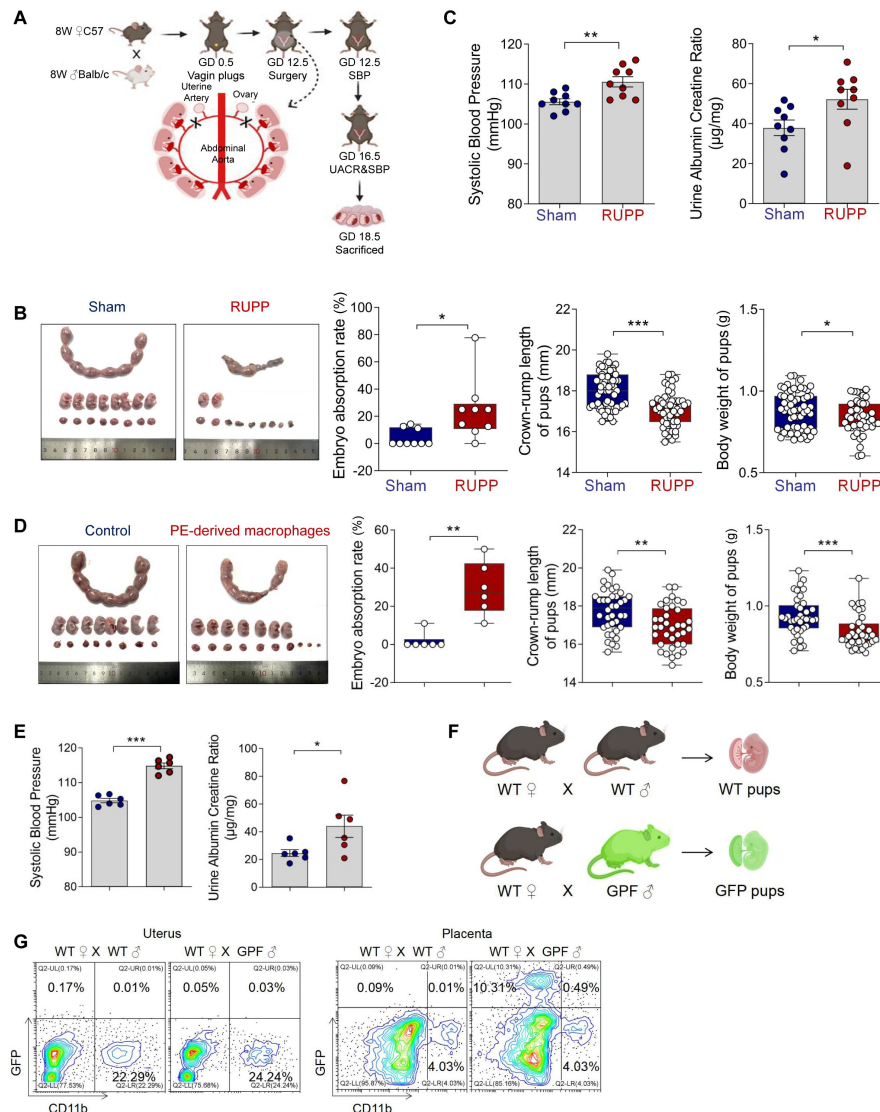


Figure 4—figure supplement 1

The construction of RUPP mouse model

(A) Experimental design of mice model with PE by reducing uterine perfusion pressure (RUPP) on day 12.5 of gestation. Mice with sham operation were considered as controls. Systolic blood pressure (SBP) and urine albumin creatine ratio (UACR) were measured on day 12.5 and 16.5 of gestation respectively. Mice were sacrificed on day 18.5 of gestation.

(B) The embryo abortion rate of pregnant mice, body weight, and crown-rump length of pups measured on day 18.5 of gestation in the Sham and RUPP group. Embryo abortion rate = number of absorbed embryos/total number of embryos. Blue represents mice in the Sham group (n=9); Red represents mice in the RUPP group (n=9).

(C) SBP and UACR of pregnant mice in Sham and RUPP group.

(D) The embryo abortion rate of pregnant mice, body weight, and crown-rump length of pups measured on day 18.5 of gestation in the mice injected with PBS or RUPP-derived macrophages. Embryo abortion rate = number of absorbed embryos/total number of embryos. Blue represents mice in the mice injected with PBS group (n=6); Red represents mice in the mice injected with RUPP-derived macrophages group (n=6).

(E) SBP and UACR of pregnant mice in PBS and RUPP-derived macrophages group.

(F) Experimental design of GFP pregnant mice model.

(G) Frequencies of GFP⁺CD11b⁺ fetal-derived macrophages and GFP⁺CD11b⁺ maternal-derived macrophages were analyzed by flow cytometry.

Data were compared between the two groups using the Student's t-test and represented as mean±SEM (*P < 0.05, ** P < 0.01, *** P < 0.001).

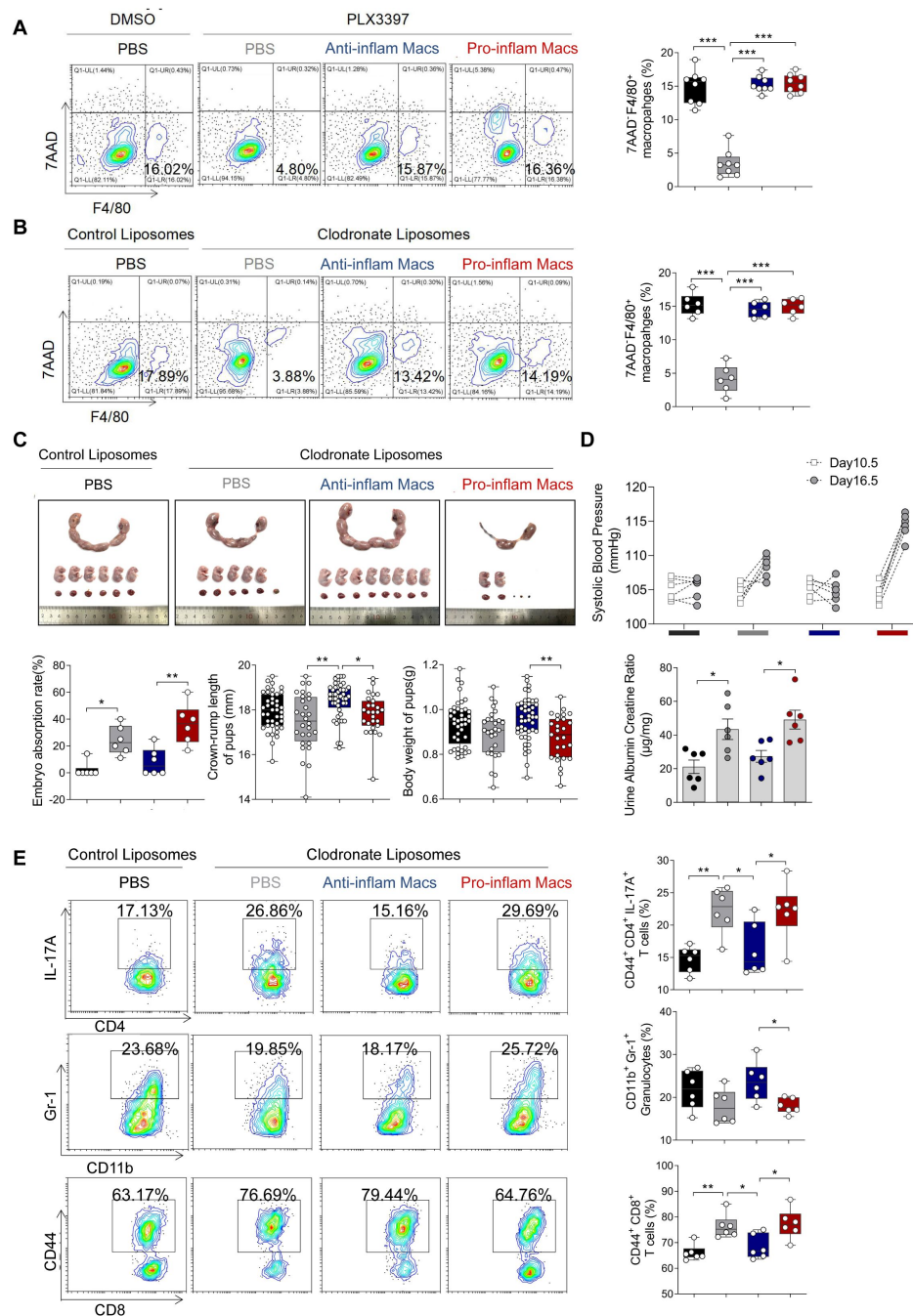


Figure 4—figure supplement 2

Clodronate liposomes were used to deplete the macrophages of pregnant mice to demonstrate that pro-inflam Macs lead to immune imbalance.

(A) Frequencies of F4/80⁺7AAD⁻ macrophages in PLX3397-treated mouse were analyzed by flow cytometry. (B) Frequencies of F4/80⁺7AAD⁻ macrophages in Clodronate liposomes-treated mouse were analyzed by flow cytometry. (C) Embryo abortion rate of the pregnant mice, body weight and crown-rump length of pups measured on day 18.5 of gestation. Black represents mice treated with control liposomes (n=6); gray represents mice treated with clodronate liposomes (n=6); blue represents mice injected with CD45⁺F4/80⁺CD206⁺ anti-inflam Macs (n=6); red represents mice injected with CD45⁺F4/80⁺CD206⁺ pro-inflam Macs (n=6). (D) SBP and UACR of pregnant mice in the four groups. (E) Frequencies of CD44⁺CD4⁺IL-17A⁺ cells, CD44⁺CD8⁺ T cells and CD11b⁺Gr1⁺ granulocytes analyzed by flow cytometry. Data were compared between groups using one-way ANOVA and represented as mean±SEM (*P < 0.05, ** P < 0.01, *** P < 0.001).

Pro-inflam Macs and anti-inflam Macs subsets are phenotypically heterogeneous

To further explore the role that macrophages play in the immune imbalance at the maternal-fetal interface in PE, scRNA-seq was performed to analyze the CD45⁺ immune cells in the uterus and placenta from mice that were injected with pro-inflam Macs or anti-inflam Macs from the uterus and placentas of RUPP mice. An unsupervised cluster detection algorithm (SEURAT) was applied and eight types of immune cells were detected by mostly distinguishable cell type-specific genes, including macrophages, monocytes, granulocytes, T/NK cells, B cells, DC-like cells, mast cells, and basophils (**Figure 5A, 5B**). Macrophages were further identified into 15 clusters defined by marker set signatures. Single-cell differential expression analysis (SCDE) was performed for each population and characteristic gene expression patterns were detected for cluster 0-cluster 14 to characterize the phenotypes of these subsets in detail (**Figure 5C**, **Figure 5-figure supplement 1C**). We found that the frequency of cluster 0 was significantly increased in mice injected with pro-inflam Macs, while the frequency of cluster 1 was significantly decreased (**Figure 5D**). We also found that cluster 0 highly expresses genes associated with fetal and tissue resident (Folr2) (Nalio et al., 2022; Thomas et al., 2021), complement system activation (C1qa, C1qb, C1qc) (Chen et al., 2021), and inflammatory response (Ccl7, Ccl8) (He et al., 2019; Wu et al., 2023); while cluster 1 highly expresses genes associated with tissue repair (Chil3, Sipi, Fn1) (Jin et al., 2023; Nugteren et al., 2021; Li et al., 2022), blood vessel morphogenesis (Thbs1) (Che et al., 2021) and preventing oxidative stress (Gsr, Mgst1) (Coppo et al., 2022) (**Figure 5E**). Changes in gene expression patterns of cluster 0 (pro-inflam Macs) and cluster 1 (anti-inflam Macs) were analyzed in mice injected the pro-inflam Macs or anti-inflam Macs. We found that enriched GO terms in cluster 0 and cluster1 including ‘antigen processing and presentation of exogenous antigen’ and ‘inflammatory response’ in the pro-inflam Macs group (**Figure 5F**), in which high expression of genes such as CCL5, CCL8, CALR, IRF7, IL10, IFI44, IFI30, OAS3, etc. could be observed (**Figure 5G**). The control group showed elevated expression of MMP14, Chil3, EGR1, ATF3 and VASP (**Figure 5G**); and these genes were enriched in ‘vascular endothelial growth factor production’ and ‘tissue development’ (**Figure 5F**). These results were consistent with the transcriptome RNA sequencing results of macrophages.

These data suggested that the CD45⁺F4/80⁺CD206⁻ pro-inflam Macs with a Folr2⁺Ccl7⁺Ccl8⁺C1qa⁺C1qb⁺C1qc⁺ phenotype play an important role in the development of PE.

Pro-inflam Macs induced the memory-like Th17 cells, which was associated with the development and recurrence of PE

We also analyzed the scRNA-seq data of uterine T/NK cells from mice that were injected with or without pro-inflam Macs. T/NK cells were further identified into 12 clusters defined by marker set signatures (**Figure 6A**, **Figure 6-figure supplement 1A**). We found that the frequency of cluster 0 was increased in mice injected with pro-inflam Macs, while the frequency of cluster 1 and cluster 2 were significantly decreased (**Figure 6B**). In addition to genes associated with Th17 cells (IL-17a, IL17f, Rora, Il1r1, TNF) (Wu et al., 2019; Leite et al., 2023), cluster 0 also exhibits high expression of genes associated with memory phenotype (Cxcr6, S100a4, CD44) (Schroeder et al., 2023; Bieberich et al., 2021), suggesting that cluster 0 was the memory-like Th17 cells (**Figure 6C**). Cluster 1 exhibited high expression of genes associated with immunoregulation (Lef1, Tcf7, Ccr7) (Qiu et al., 2022; Sekine et al., 2020); Cluster 2 exhibited high expression of genes associated with immunomodulation (Gata3, Gitr, CD28), suggesting that cluster 1 and 2 were T cells with immunomodulation function (Esensten et al., 2016; Pai et al., 2023) (**Figure 6-figure supplement 1B**). Changes in gene expression patterns of cluster 0 (memory-like Th17 cells) were analyzed in the pro-inflam Macs group and control group. We found that in the pro-inflam Macs group, enriched GO terms including ‘positive regulation of response to external stimulus’ and ‘tumor necrosis factor production’ (**Figure 6D**), in which

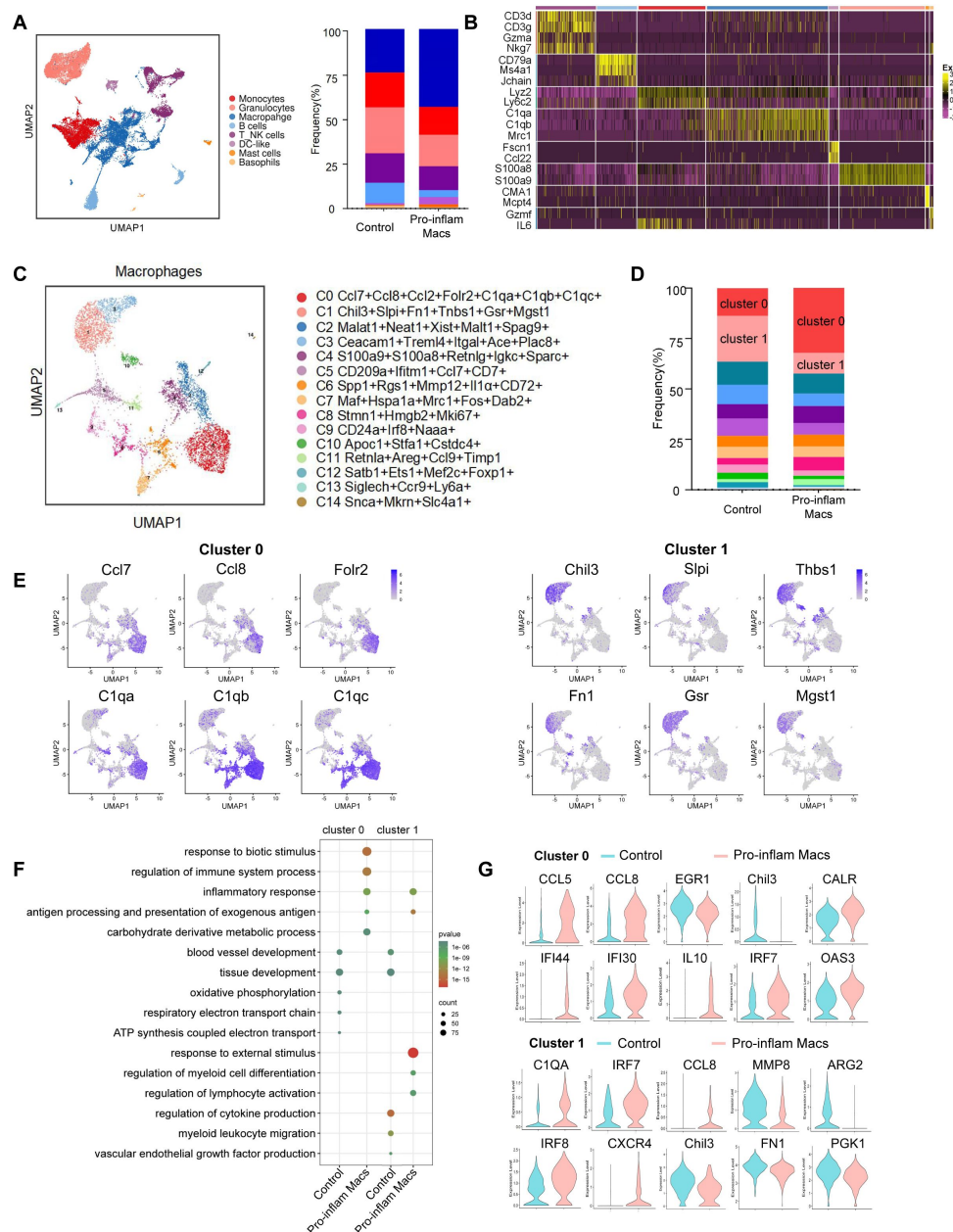


Figure 5

Identification of the macrophage subsets in mice injected with pro-inflam Macs and anti-inflam Macs by using scRNA-seq.

- (A) UMAP maps showing the 8 types of mouse immune cells at the maternal-fetal interface.
- (B) Heatmap showing clustering analysis for markers distinguished different type of immune cells.
- (C) UMAP maps showing the 15 clusters of mouse macrophages was listed.
- (D) Bar graph showing the frequencies of clusters of macrophages in the two groups of mice was listed in the right panel.
- (E) UMAP maps showing the distribution of specific markers of cluster 0 and cluster 1.
- (F) Dot plot depicting GO enrichment terms that were significantly enriched in the differentially expressed genes in cluster 0 and cluster 1 from the pro-inflam Macs group and the control group.
- (G) Violin plot of specific differential gene expression in cluster 0 and cluster 1 between the pro-inflam Macs group and the control group.

Data were compared between the two groups using the Student's t-test and represented as mean $\pm$ SEM (*P < 0.05, ** P < 0.01, *** P < 0.001).

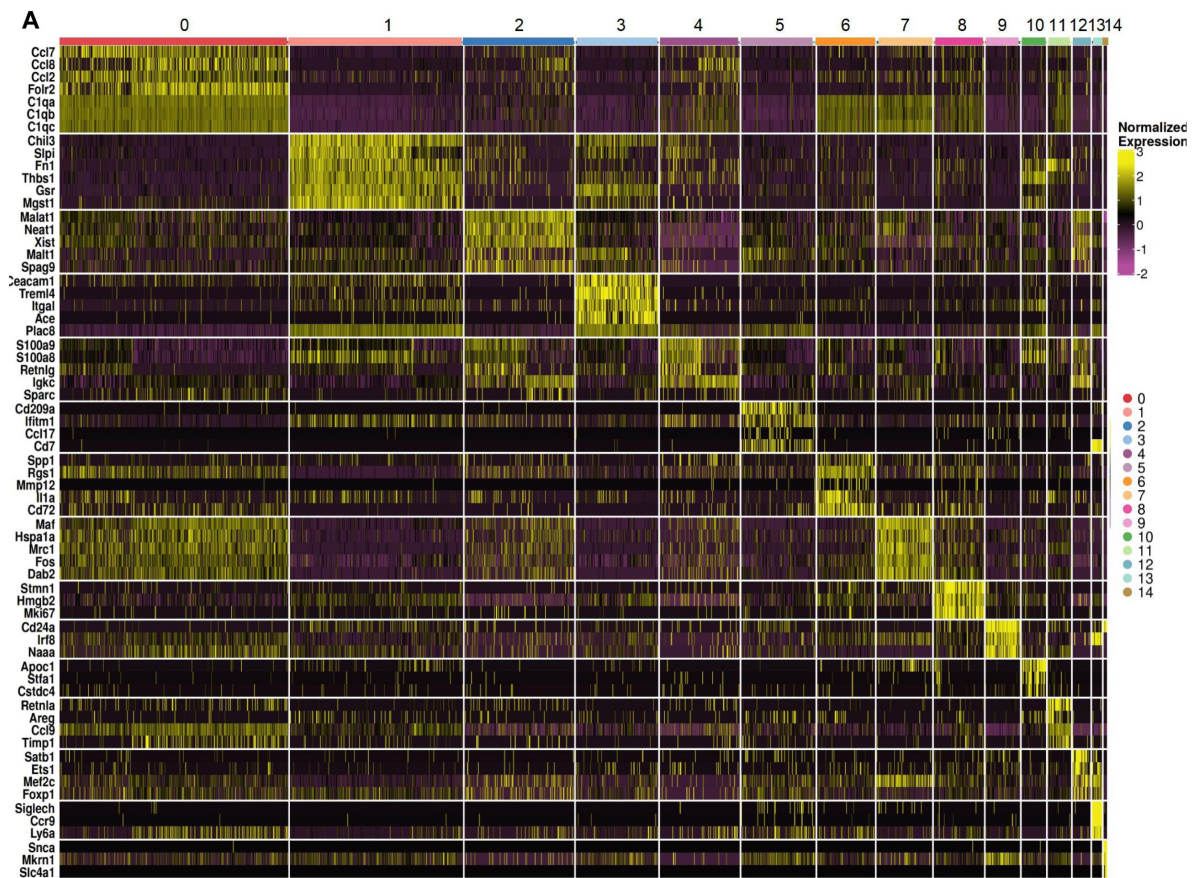


Figure 5—figure supplement 1

Heatmap of different clusters of macrophages.

(A) Heatmap showing clustering analysis for markers distinguished different clusters of macrophages.

highly express genes such as TNFSF11, TNF, IL27RA, IGF1R, CD226, LAMP1 (**Figure 6E**). The control group showed elevated expression of ASS1, EIF5, S100A9, CTLA4, S100A8 and CXCR4 (**Figure 6E**); these genes were enriched in GO terms including ‘regulation of programmed cell death’ and ‘cellular biosynthetic process’ (**Figure 6D**). In summary, the pro-inflam Macs induced memory-like Th17 cells, which were characterized by IL-17a⁺IL17f⁺Rora⁺Il1r1⁺TNF⁺Cxcr6⁺S100a4⁺CD44⁺.

To confirm the importance of memory-like CD4⁺ T cells in the pathogenesis of PE, the CD4⁺CD44⁺ T cells in the uterus and placentas from RUPP or NP mice were sorted and intravenously injected into normal pregnant mice on day12.5 of gestation (**Figure 6F**). The CD4⁺CD44⁺ T cells derived from RUPP mouse were characterized by an increased frequency of Th17 cells and a reduced frequency of Tregs (**Figure 6F**). Increased embryo absorption rate, decreased fetal weight and pup crown-rump length were found in mice injected with PE mouse-derived CD4⁺CD44⁺ T cells (**Figure 6G**). The PE-like symptoms, including increased SBP and UACR, were also observed in those mice (**Figure 6H**).

It has been reported that sustained expansion of immunosuppressive memory Tregs during prior pregnancy is beneficial for maintaining a second pregnancy (Rowe et al., 2012). To verify whether memory-like Th17 cells promote the recurrence of PE, we established a second pregnant mouse model with a history of PE or NP in the first pregnancy. An increased embryo resorption rate, decreased crown-rump length, and fetal weight were found in mice with a history of PE pregnancy compared with those with a history of NP pregnancy (**Figure 6-figure supplement 1C**). Moreover, mice with a history of PE in the first pregnancy showed increased SBP and UACR during the second pregnancy (**Figure 6-figure supplement 1D**). Consistently, higher levels of IL-17A were also found in memory-like CD4⁺ T cells in mice with a history of PE in the first pregnancy (**Figure 6-figure supplement 1E**).

The uterine granulocytes from mice that were transferred with pro-inflam Macs from PE mice were also analyzed using scRNA-seq. Granulocytes were identified into 12 clusters defined by marker set signatures (**Figure 6-figure supplement 2A, 2B**). We found that the frequency of cluster 3 were decreased (**Figure 6-figure supplement 2A**). And cluster 3 highly expressed gMDSCs associated genes (Ly6g, S100a8, Retnlg, Wfdc21) (von Wulffen et al., 2023; Kao et al., 2023) (**Figure 6-figure supplement 2C**). Changes in gene expression patterns of cluster 3 (gMDSCs) were analyzed in the pro-inflam Macs group and the control group. Enriched GO terms including ‘tumor necrosis factor production’ and ‘leukocyte mediated immunity’ were found in the pro-inflam Macs group (**Figure 6-figure supplement 2E**), which highly express the genes such as IRF7, EGR1, C1QA, C1QB, CCL2, CSF3R, TNFRSF1B (**Figure 6-figure supplement 2D**). Except for gMDSCs, we also revealed a pronounced increase in the proportion of CD15⁺CD66b⁺ neutrophils among PE patients, implying the importance of neutrophils in the development of PE (Akkari L et al., 2024) (**Figure 6-figure supplement 2F**).

In conclusion, pro-inflam Macs could induced the differentiation of IL-17a⁺IL17f⁺Rora⁺Il1r1⁺TNF⁺Cxcr6⁺S100a4⁺CD44⁺ memory-like Th17 cells, which was associated with the development and recurrence of PE. And the pro-inflam Macs also suppressed the production of Ly6g⁺S100a8⁺Retnlg⁺Wfdc21⁺ gMDSCs.

Pro-inflam Macs induced the production of memory-like Th17 cells via IGF1-IGF1R

Our results above showed that pro-inflam Macs induced memory-like Th17 cells in PE, however, the underlying molecular mechanisms were still unknown. CellPhoneDB analysis indicated that increased communication counts and signaling pathways numbers between macrophages and T/NK cells were observed in mice injected with pro-inflam Macs (**Figure 7—figure supplement 1A, 1B**). Then we identified the interacting ligand-receptor pairs between different types of immune

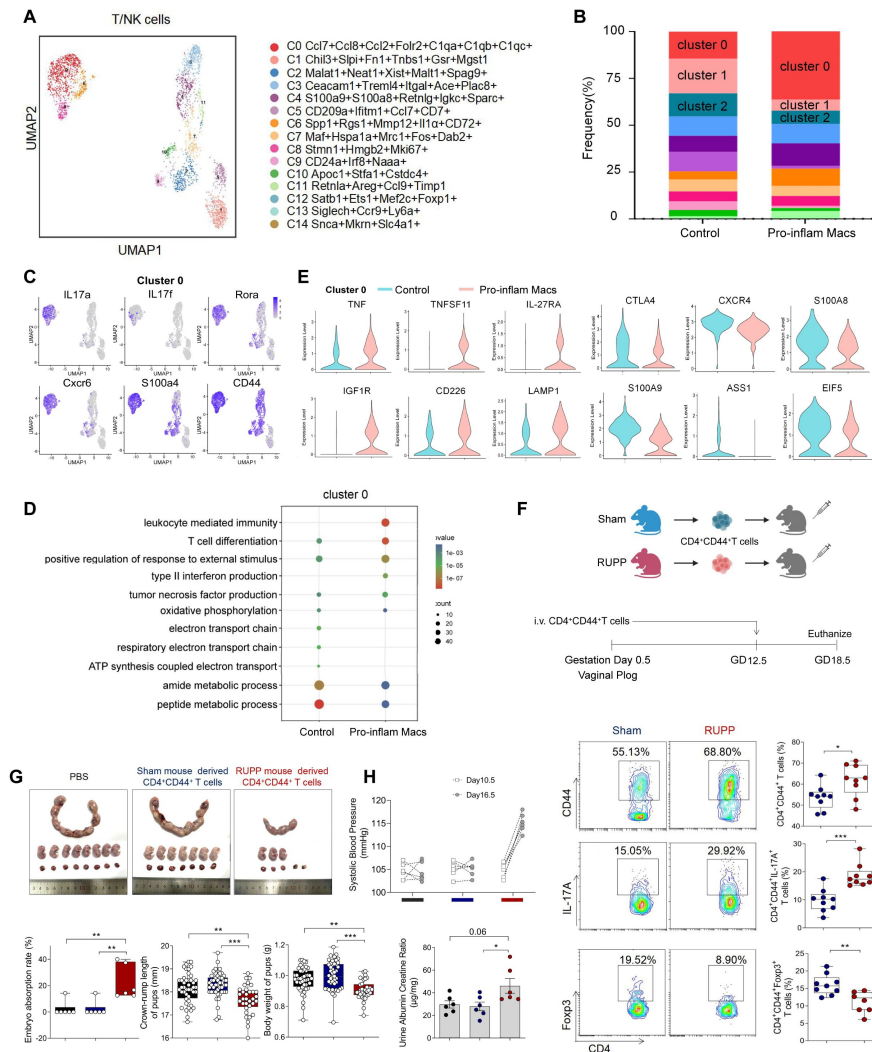


Figure 6

Identification of the T/NK cells subsets in mice injected pro-inflam Macs and anti-inflam Macs using scRNA-seq.

(A) UMAP maps showing the 12 clusters of mouse T/NK cells was listed. Heatmap showing clustering analysis for markers distinguished 12 different clusters of T/NK cells.

(B) Bar graph showing the frequencies of clusters of T/NK cells in the two groups of mice was listed in the down panel.

(C) Dot plot depicting GO enrichment terms that were significantly enriched in the differentially expressed genes in cluster 0 from the pro-inflam Macs group and the control group.

(D) Violin plot of specific differential gene expression in cluster 0 between the pro-inflam Macs group and the control group.

(F) Frequencies of CD4⁺CD44⁺ T cells and the percentages of Foxp3⁺ or IL-17A⁺ cells in CD4⁺CD44⁺ T cells at the maternal-fetal interface in Sham and RUPP group analyzed by flow cytometry.

(G) The embryo abortion rate of pregnant mice, body weight, and crown-rump length of pups measured on day 18.5 of gestation in mice injected PBS, Sham mouse-derived or RUPP mouse-derived CD4⁺CD44⁺ T cells. Black represents mice injected with PBS (n=6); blue represents mice injected with Sham mouse-derived CD4⁺CD44⁺ T cells (n=6); red represents mice injected with RUPP mouse-derived CD4⁺CD44⁺ T cells (n=6).

(H) SBP and UACR of pregnant mice injected with PBS, Sham mouse-derived or RUPP mouse-derived CD4⁺CD44⁺ T cells. Data were compared between groups using one-way ANOVA and represented as mean±SEM (*P < 0.05, ** P < 0.01, *** P < 0.001).

Data were compared between the two groups using the Student's t-test and represented as mean±SEM (*P < 0.05, ** P < 0.01, *** P < 0.001).

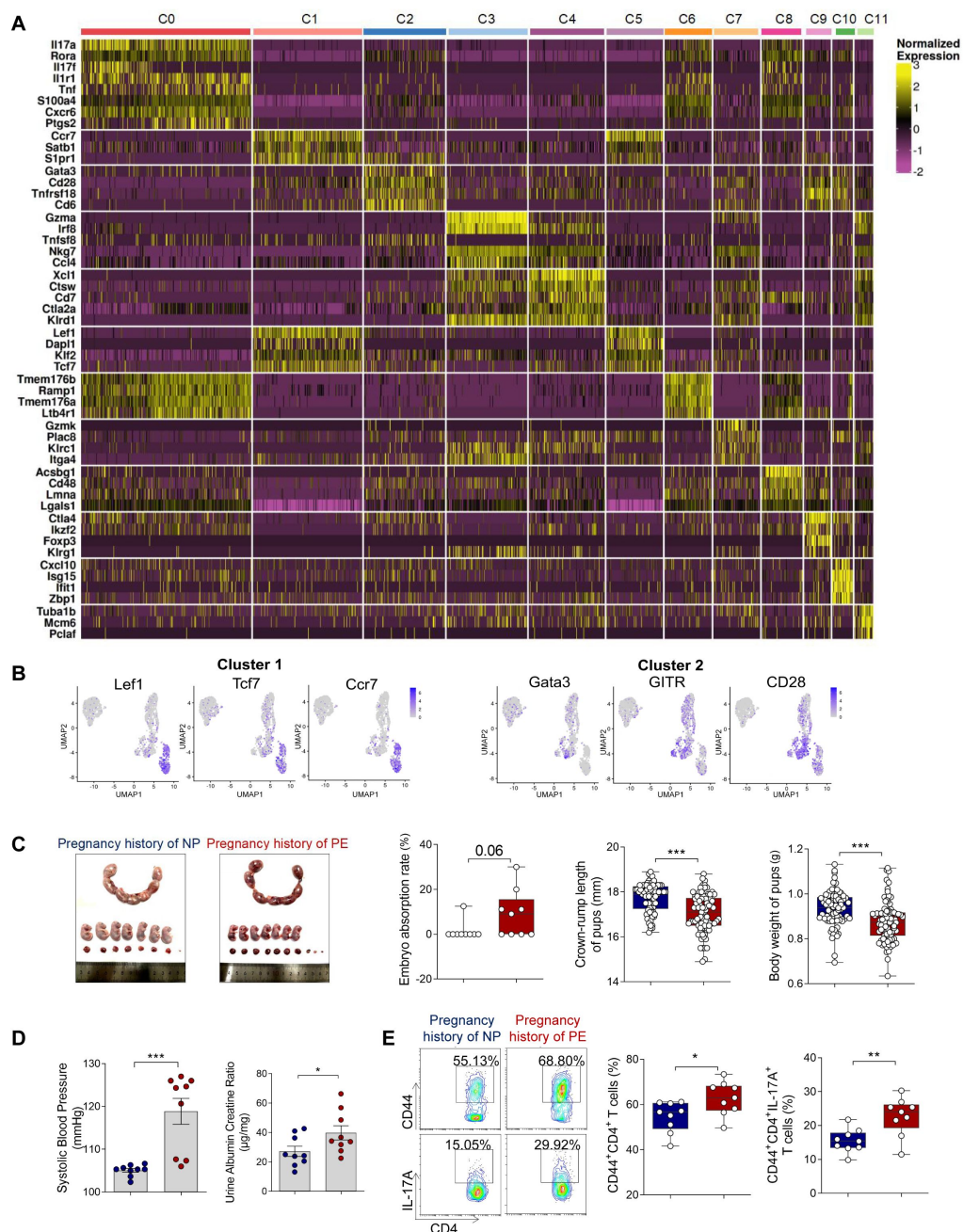


Figure 6—figure supplement 1

Memory-like Th17 cells may be associated with the recurrence of PE.

(A) Heatmap showing clustering analysis for markers distinguished 12 different clusters of T/NK cells.

(B) Embryo abortion rate of pregnant mice, body weight and crown-rump length of pups measured on day 18.5 of gestation. Blue represents mice with previous normal pregnancy (n=9); red represents mice with previous pregnancy with PE (n=9).

(C) SBP and UACR of second pregnant mice with a history of PE or NP in the first pregnancy measured on day 16.5 of gestation.

(D) Frequencies of CD4⁺ CD44⁺ T cells and the levels of IL-17A in CD4⁺ CD44⁺ T cells in mice with a pregnancy history with NP and PE analyzed by flow cytometry.

Data were compared using the Student's t-test and represented as mean±SEM (*P < 0.05, ** P < 0.01, *** P < 0.001; NS, not significant).

Data were compared between the two groups using the Student's t-test and represented as mean±SEM (*P < 0.05, ** P < 0.01, *** P < 0.001).

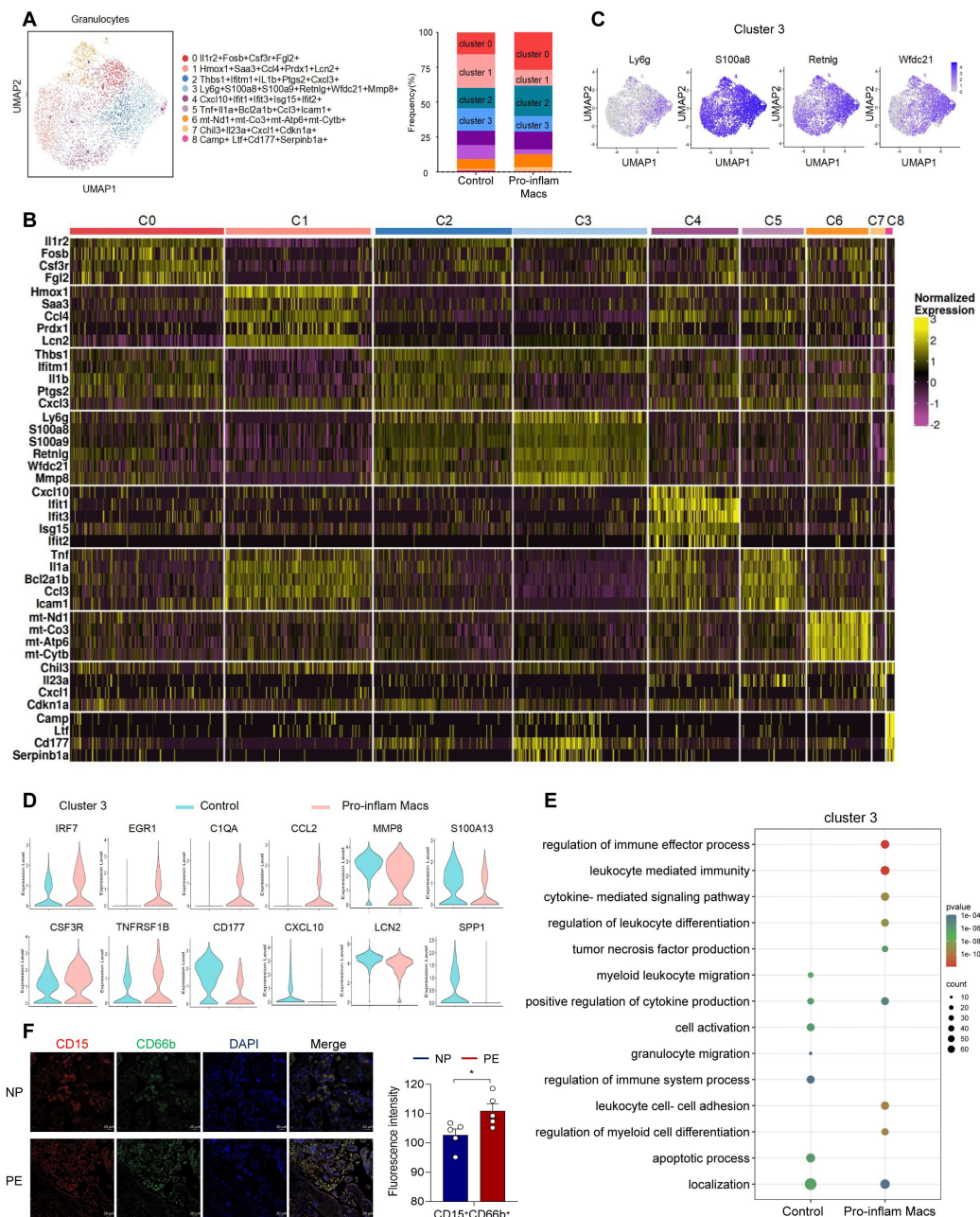


Figure 6—figure supplement 2

Identification of the granulocytes subsets in mice injected pro-inflam Macs and anti-inflam Macs use sc RNA-seq.

(A) UMAP maps showing the 9 clusters of mouse granulocytes. Bar graph showing the frequencies of clusters of granulocytes in the two groups of mice.

(B) Heatmap showing clustering analysis for markers distinguished 9 different clusters of granulocytes cells.

(C) UMAP maps showing the distribution of specific markers of cluster 3.

(D) Dot plot depicting GO enrichment terms that were significantly enriched in the differentially expressed genes in cluster 3 from the pro-inflam Macs group and the control group.

(E) Violin plot of specific differential gene expression in cluster 3 between the pro-inflam Macs group and the control group.

(F) Immunofluorescence co-staining of CD15 (red), CD66b (green) and DAPI (blue) in frozen placental sections. Scale bar, 20 μ m.

Data were compared between the two groups using the Student's t-test and represented as mean $\pm$ SEM (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$).

cells and macrophages. Insulin-like growth factor 1 (IGF1) -IGF1 receptor (IGF1R) ligand-receptor pair was significantly enhanced between macrophages and T/NK cells in mice injected with pro-inflam Macs (**Figure 7A**). IGF1 has significant effects on immune function maintenance, and signaling through IGF1R could cause increased aerobic glycolysis, favoring Th17 cell differentiation over that of Treg cells (Bekkering et al., 2018; DiToro et al., 2020). For further demonstration, we analyzed the frequencies of IGF1⁺CD14⁺ cells and IGF1R⁺CD4⁺ cells in placentas of individuals with PE and NP and found both of these two cells were significantly increased in the PE group (**Figure 7B**). Trophoblast-derived extracellular vesicles from the placenta of PE (PE-EVs), which carry the fetal antigen, could induce M1-like macrophage (pro-inflammatory Macs) polarization to participate in the development of PE according to our previous study (X. Liu et al., 2022). To confirm whether pro-inflammatory Macs induced the production of memory-like Th17 cells via IGF1-IGF1R, macrophages from human peripheral blood, after incubating with PBS, trophoblast-derived extracellular vesicles from NP (NP-EVs) or PE-EVs in vitro, were co-cultured with CD4⁺ naïve T cells (**Figure 7C**). The frequencies of memory-like CD45RO⁺ CCR7⁺ IL-17A⁺ CD4⁺ cells significantly increased in the PE-EVs-induced macrophages group rather than in the NP-EVs-induced macrophages group. However, the frequencies of memory-like Th17 cells significantly decreased after CD4⁺ naïve T cells were treated with the IGF1R inhibitor BMS-754807 (**Figure 7D**).

To investigate the role of IGF1-IGF1R in the development of PE in vivo, CD45⁺CD4⁺ T cells from NP mice were isolated and treated with BMS-754807 or PBS, then injected into NP mice at day 11.5 of gestation (**Figure 7E**). Lipopolysaccharide (LPS)-induced PE mice model was constructed by intraperitoneal injection of LPS on day12.5 and day15.5 of gestation (Han et al., 2021), and an anti-CD4 antibody was used to deplete the CD4⁺ T cells in pregnant mice. The mice injected with CD4⁺ T cells treated by BMS-754807 presented decreased embryo resorption rate and increased pup crown-rump length and fetal weight (**Figure 7F**), decreased SBP and UACR (**Figure 7G**), and decreased frequency of memory-like Th17 cells at the maternal-fetal interface (**Figure 7H**). These data suggested that pro-inflam Macs could induce the production of memory-like Th17 via the IGF1-IGF1R, leading to the development of PE.

Discussion

Preeclampsia, a progressive systemic disease during pregnancy, is closely related to the alterations in the immune environment at the maternal-fetal interface. In this study, the overall immune cell profiles in the placenta of individuals with NP, PE, GDM and GDM&PE were detected by CyTOF and a PE-specific change in immune microenvironment of maternal-fetal interface was described for the first time. Further, we provided a novel insight that pro-inflam Macs induced memory-like Th17 cells, memory-like CD8⁺ T cells, and suppressed the production of gMDSCs in mice maternal-fetal interface by scRNA-seq. In addition, we first validated that IGF1-IGF1R was involved in producing memory-like Th17 cells induced by pro-inflam Macs in vitro and in vivo, thus leading to the development of PE.

Various studies represented the effect of a single subset of immune cells in the pathogenesis of PE or GDM (Bachmayer et al., 2006; Care et al., 2018; Cornelius et al., 2015a; Eghbal-Fard et al., 2019; Faas et al., 2014; Fu et al., 2014; Fukui et al., 2011; Lampé et al., 2015; Lampé et al., 2011; Lang et al., 2021; D. Liu et al., 2021; Lu et al., 2020; Santner-Nanan et al., 2009; Sasaki et al., 2007; Travis et al., 2020; Yao et al., 2019). Miller et al. provided an overall analysis of immune cells in the human placental villi in the presence and absence of spontaneous labor at term by scRNA-seq (Miller et al., 2022). Fangyuan Luo et al. unveiled the intricate interplay existing between SPP1⁺M1 cells and the IGFBP1⁺SPP1⁺ extracellular villous trophoblast in the context of PE by using sc-RNA seq (Luo F et al., 2023). Li Y et al. highlighted a key role of a distinct perivascular inflammatory CD11c^{high} decidual macrophages subpopulation in the pathogenesis of preeclampsia through sc-RNA sequencing (Li Y et al., 2024). However, The

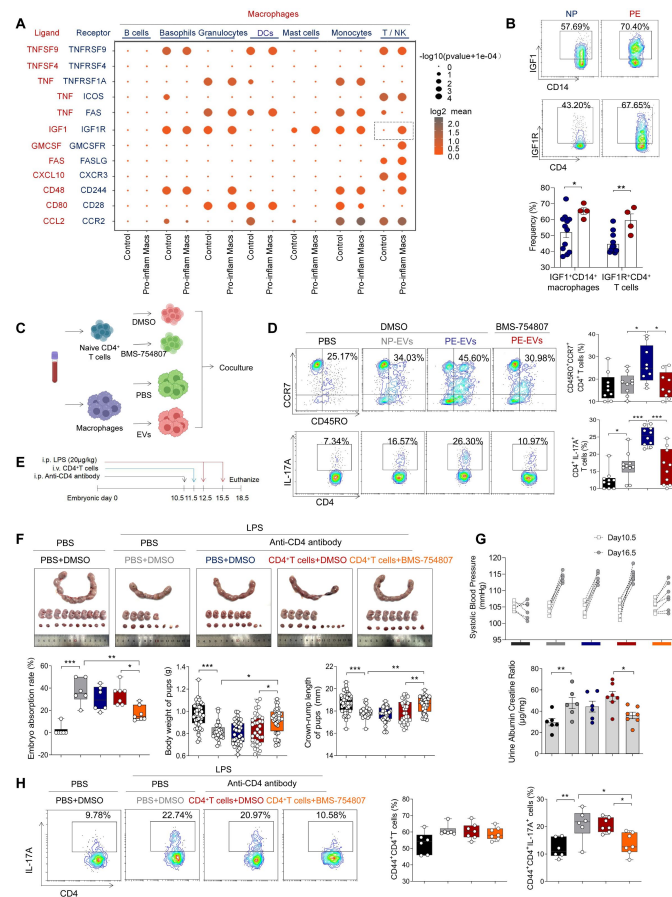


Figure 7

Pro-inflam Macs induce the generation of memory-like Th17 cells via IGF1-IGF1R

(A) Signaling modules indicated by ligand-receptor pairing between macrophages and other types of immune cells at the maternal-fetal interface using CellPhoneDB.

(B) Frequencies of IGF1⁺CD14⁺ cells and IGF1R⁺CD4⁺ cells in placentas of individuals with NP and PE (n=12 in NP group, n=4 in PE group).

(C) Schematic of the experimental workflow to induce memory-like T cells in vitro. Macrophages, after incubating with PBS, NP-EVs or PE-EVs, were co-cultured with CD4⁺ naïve T cells treated with DMSO or BMS-754807. Cells were isolated from human peripheral blood.

(D) Frequencies of CD45RO⁺CCR7⁺Th17 cells. Black represents CD4⁺ naïve T cells treated with DMSO; gray represents CD4⁺ naïve T cells cocultured with NP-EVs-treated macrophages; blue represents CD4⁺ naïve T cells co-cultured with PE-EVs-treated macrophages; red represents CD4⁺ naïve T cells treated with BMS-754807 before co-cultured with PE-EVs-treated macrophages (n=10 in each group).

(E) Schematic of mice transferred CD4⁺ T cells treated with BMS-754807 or PBS. Anti-CD4 antibody was used to deplete CD4⁺ T cells in mice on day 10.5 of gestation. CD4⁺ T cells were transferred into mice on day 11.5 of gestation. 20 μg/kg lipopolysaccharide (LPS) was intraperitoneally injected on day 12.5 and 15.5 of gestation to induce a PE-like pregnant mice model. Mice were sacrificed on day 18.5 of gestation.

(F) Embryo abortion rate of pregnant mice, body weight and crown-rump length of pups were measured on day 18.5 of gestation. Black represents the control group mice (n=6); gray represents mice treated with LPS (20μg/kg) to construct an animal model of PE (n=6); blue represents anti-CD4 antibody treated PE mice (n=6); red represents anti-CD4 antibody treated PE mice injected with CD4⁺ T cells with DMSO treatment (n=7); orange represents anti-CD4 antibody treated PE mice injected with CD4⁺ T cells with BMS754807 treatment (n=7).

(G) SBP and UACR of pregnant mice in the five groups.

(H) The frequencies of CD4⁺CD44⁺IL-17A⁺ cells analyzed by flow cytometry.

Data were compared between groups using one-way ANOVA and represented as mean±SEM (*P < 0.05, ** P < 0.01, *** P < 0.001).

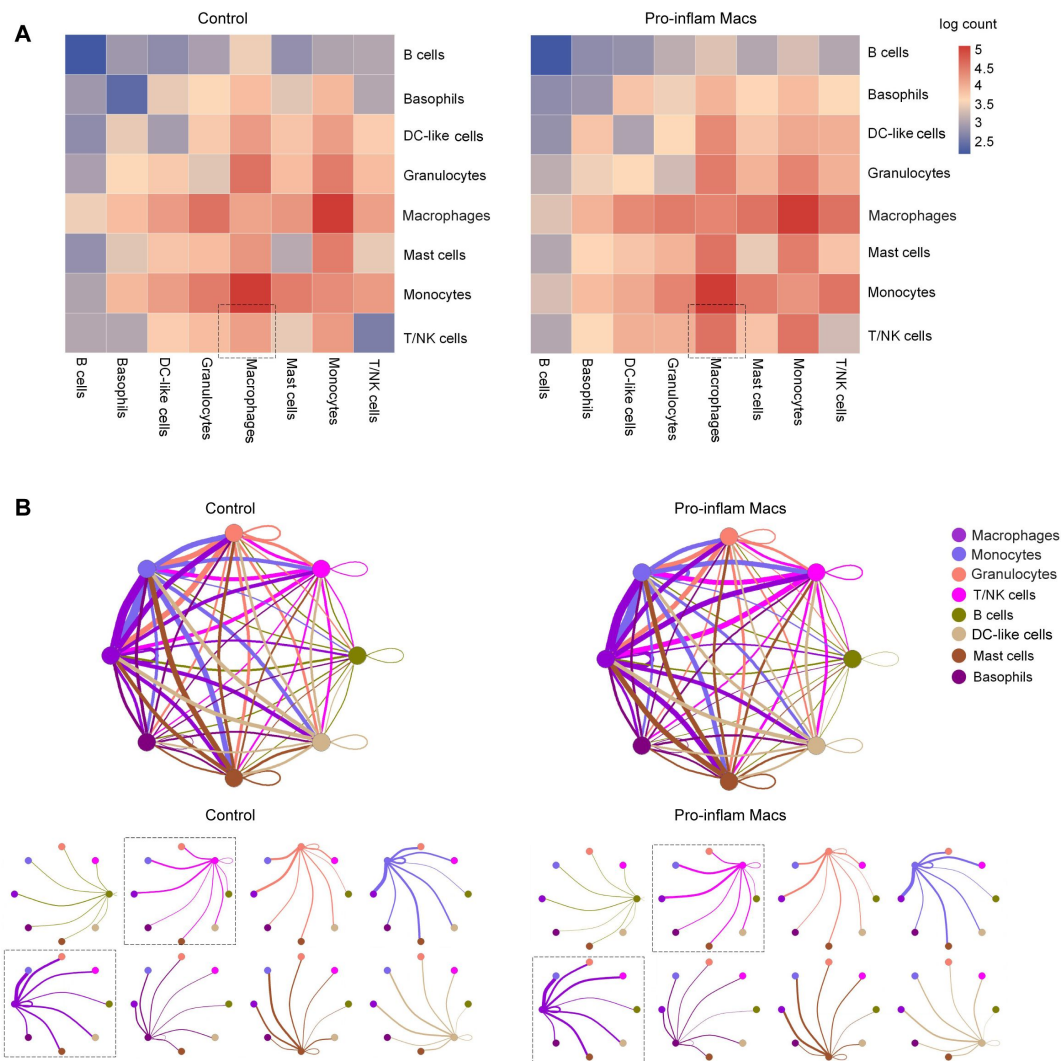


Figure 7—figure supplement 1

Cell-cell communications in immune cells when pro-inflam Macs accumulated at the maternal-fetal interface.

(A) Abundance of connections between different cell types at the maternal-fetal interface analyzed by CellPhoneDB. (B) Capacities for interactions between immune cells are showed. Each line indicates the ligands expressed by the cell population represented by the same color (labeled). The lines connect to cell types that express cognate receptors. Line thickness is proportional to the number of ligands when cognate receptors are present in the recipient cell type.

immune cells crosstalk with each other at the maternal-fetal interface elaborately. A comprehensive and in-depth understanding of the immune microenvironment at the maternal-fetal interface of PE is still lacking. Here, we used CyTOF and scRNA-seq to comprehensively analyze the immune cell profile and explore the interaction of immune cells in PE. Additionally, most previous studies have reported immune cells from maternal systemic circulation rather than directly from the tissue of maternal-fetal interface (Cornelius et al., 2015b [↗](#); Deer et al., 2021 [↗](#); Santner-Nanan et al., 2009 [↗](#); Shields et al., 2018 [↗](#); Wallace et al., 2012 [↗](#)). In this study, cells were directly isolated from the tissue of maternal-fetal interface, which is considered as a more accurate approach to unveil the real immune environment of the maternal-fetal interface.

Abnormal inflammatory responses were reported to play an important role in the pathogenesis of both PE and GDM (Aneman et al., 2020 [↗](#); Corrêa-Silva et al., 2018 [↗](#); Deer et al., 2023 [↗](#); Jung et al., 2022 [↗](#); McElwain et al., 2021 [↗](#)). However, it is unclear how the maternal-fetal interface immune microenvironment differs between the two diseases. We analyzed the overall placental immune cell profile in NP, PE, GDM and GDM&PE by CyTOF and revealed a PE-specific immune cell profile: The frequencies of memory-like Th17 cells (CD45RA⁺CCR7⁺IL-17A⁺CD4⁺), memory-like CD8⁺ T cells (CD45RA⁺CCR7⁺CD38⁺pAKT^{mid}CD127^{low}CD8⁺) and pro-inflam Macs (CD206⁺CD163⁺CD38^{mid}CD107a^{low}CD86^{mid}HLA-DR^{mid}CD14⁺) were increased, while the decreased frequencies of CD69^{hi}Helios^{mid}CD127^{mid}γδT cells, anti-inflam Macs (CD206⁺CD163⁺CD86^{mid}CD33⁺HLA-DR⁺CD14⁺) and granulocyte myeloid-derived suppressor cells (gMDSCs, CD11b⁺CD15^{hi}HLA-DR^{low}) were observed in the placenta of PE, but not in that of GDM or GDM&PE compared with that of NP.

PE is often divided into two subtypes based on the time of onset. Late-onset PE, occurring at 34 weeks of gestation or later, accounts for most preeclampsia cases, while early-onset PE, occurring before or at 33 weeks of gestation (Lisonkova and Joseph, 2013 [↗](#)). However, our previous study observed no significant difference in the M1/M2 macrophage polarization in placentas with early-onset or late-onset PE (X. Liu et al., 2022 [↗](#)). Therefore, our research was conducted regardless of the preeclampsia-onset type to ensure consistency in macrophage polarization of the two subtypes.

There have been studies indicating that pro-inflammatory macrophages, Th17 cells and CD8⁺ T cells play an essential role in PE development (Care et al., 2018 [↗](#); Eghbal-Fard et al., 2019 [↗](#); Lager et al., 2020 [↗](#); Yao et al., 2019 [↗](#)). Macrophages was indicated with a pro-inflammatory phenotype in the placenta with PE (Faas et al., 2014 [↗](#); Yao et al., 2019 [↗](#)). Care et al. reported that placental Tregs helped suppress inflammation at the maternal-fetal interface (Care et al., 2018 [↗](#)). Similarly, Lu et al. found an abnormal increase of IL-17A in the placenta in the development of PE (Lu et al., 2020 [↗](#)). Meanwhile, the abnormal infiltration of CD8⁺ T cells in the placenta may contribute to PE (Lager et al., 2020 [↗](#)). Consistent with the previous findings, we found that memory-like CD4⁺ and memory-like CD8⁺ T cells significantly increased in the PE group and that IL-17A was significantly higher expressed in CD4⁺ memory-like T cells from the PE group, which is vital in vascular inflammation, leading to the development of PE (Amador et al., 2014 [↗](#); Madhur et al., 2021 [↗](#)). Though memory T cells play a key role in fetal-maternal tolerance in normal pregnancy (Kieffer et al., 2019 [↗](#)), few studies have reported the characteristics of memory-like T cells in the development of PE.

We constructed an adoptive transferred mouse model. The injection of CD4⁺ T cells from RUPP mouse, characterized by an increased frequency of Th17 cells and a reduced frequency of Tregs, induced PE-like symptoms in pregnant mice. Tregs which are well known to undergird maternal tolerance of the fetus, and which are well known to have an overlapping developmental trajectory with RORγt⁺ Th17 cells (Robertson SA et al., 2019 [↗](#); Sefik E et al., 2015 [↗](#)). This study underscores the importance of Th17/Treg cell balance in maintaining pregnancy homeostasis. And it is worth noting that we isolated CD4⁺ memory-like T cells directly from the maternal-fetal interface, which is different from previous studies that the CD4⁺ T cells for adoptive transfer were isolated from the spleen and induced in vitro (Cornelius et al., 2015b [↗](#); Deer et al., 2021 [↗](#); Shields et al., 2018 [↗](#);

Wallace et al., 2012 [↗](#)). Recurrence is an important concern in individuals with PE (Bernardes et al., 2019 [↗](#)). We found that a prior PE pregnancy increased the probability of PE in the next pregnancy in mice, accompanied by an increased frequency of memory-like Th17 cells at the maternal-fetal interface, suggesting that increased memory-like Th17 cells are positively related to the recurrence of PE. However, further studies are needed to confirm the function of the fetal-specific memory Th17 cells in the recurrence of PE.

Activation of CD8⁺ T cells and Th1 cells can be induced by abnormal polarized macrophages in recurrent miscarriage (Li et al., 2022 [↗](#)). The gMDSCs have been reported to have immune suppressive activity and are necessary for maintaining maternal-fetal tolerance (Köstlin-Gille et al., 2019 [↗](#); Zhou et al., 2018 [↗](#)). However, the correlation between macrophages and other immune cells in the development of PE remained unclear. We verified that pro-inflam Macs induced memory-like Th17 cells and memory-like CD8⁺ T cells and inhibited the production of gMDSCs in PE by correlation analysis and adoptive transfer in animal experiments for the first time, indicating that the immune cells orchestrate a network in the maternal-fetal interface elaborately. However, PLX3397, used in this study, emerges as a promising small-molecule compound, endowed with the capability to traverse the placental barrier, thereby potentially influencing the development and function of fetal macrophages. In addition to gMDSCs, we conducted immunofluorescence analysis of placental tissue, specifically targeting CD66b^{high} neutrophils. Our findings revealed a pronounced increase in the proportion of neutrophils among PE patients, fostering the hypothesis that IL-17A produced by Th17 cells might orchestrate the migration of neutrophils towards the placental milieu. This migratory response, in turn, could potentially trigger reactive oxygen species (ROS)-mediated immunopathology, which subsequently lead to PE (Akkari L et al., 2024 [↗](#)).

The function of IGF1 and IGF1R in PE is controversial. IGF1 plays a significant role in maintaining immune function, and IGF1R facilitates the differentiation of naïve CD4⁺ T cells into Th17 cells (DiToro et al., 2020 [↗](#)). A study showed that IGF-1R was downregulated in the serum of individuals with PE (Liao et al., 2021 [↗](#)). Decreased amounts of IGF1R were also found in the placentas of individuals with PE (Robajac et al., 2015). However, another study found no significance in the affinity and the number of IGF-1R between placentas from individuals with PE or not (Díaz et al., 2005 [↗](#)). We found that mice exhibiting PE symptoms had significantly enhanced IGF1-IGF1R interaction between inflammatory macrophages and memory-like Th17 cells by analyzing the data of scRNA-seq. Then, this finding was further confirmed in human placentas from individuals with PE and NP, for the frequencies of IGF1⁺ CD14⁺ cells and IGF1R⁺ CD4⁺ cells were found to significantly increase in the PE group. In vitro, we found that PE-EVs-treated macrophages secreted more IGF1 than NP-EV-treated macrophages and induced a higher frequency of memory-like Th17 cells. The inhibition of IGF1R on CD4⁺ naïve T cells resulted in a decreased frequency of memory-like Th17 cells, implying that the IGF1-IGF1R is critical for the production of memory-like Th17 cells induced by inflammatory macrophages. Animal experiments confirmed that IGF1R inhibition on CD4⁺ T cells at the maternal-fetal interface reduced the frequency of memory-like Th17 cells and relieved PE-like symptoms.

In addition, placentas from different groups for CyTOF analysis and flow cytometry were matched by age, preterm pregnancies, and previous abortion history; however, there are differences in body mass index (BMI), gestational age, term pregnancies, and the number of living children. Obesity can increase inflammatory and oxidative stress markers in the placental environment (Spradley et al., 2015 [↗](#)). Also, it has been reported that the placental immune state shifts with gestational age (Lewis et al., 2018 [↗](#)). However, as PE is often accompanied by obesity and early termination of pregnancy, it is difficult to exclude these factors in sample collection. Moreover, limited placental samples in the GDM&PE group are the shortage of this study, for it is hard to collect enough clean samples that exclude interference factors because the number of pregnant women exposed to COVID-19 has increased sharply since December 2022 in China.

In summary, this study demonstrated the statistically distinguished placental immune microenvironment in individuals with PE, but not in those with GDM or GDM&PE. More importantly, macrophages orchestrate a network in the maternal-fetal interface elaborately. These findings provide novel insights into PE pathogenesis and a potential immune target for the clinical prevention and treatment of PE.

Materials and Methods Clinical sample collection

The samples used in this study were collected from Sir Run Run Shaw Hospital between October 2020 and August 2023. Informed consent was obtained from all volunteers, and the Ethics Committee of Sir Run Run Shaw Hospital, Zhejiang University School of Medicine, approved the study. Human placentas were obtained from women with NP, PE, GDM, or GDM&PE who underwent elective cesarean delivery. The diagnostic criteria for PE included new-onset hypertension after 20 weeks of gestation with SBP ≥ 140 mmHg and/or diastolic blood pressure ≥ 90 mmHg and proteinuria (≥ 300 mg) on at least two occasions. A positive glucose tolerance test diagnoses gestational diabetes mellitus. Women with normal blood pressure, full-term pregnancies, and no complications were designated as controls. The detailed clinical characteristics of the pregnant women in this study are presented in [Table 5](#) and [Table 6](#).

Mice

Eight-week-old female C57 mice and ten-week-old male BALB/c mice were purchased from Hangzhou Ziyuan Laboratory Animal Technology Co., Ltd (Hangzhou, China) and Shanghai Jihui Experimental Animal Breeding Co., Ltd (Shanghai, China), respectively. All animals were maintained under pathogen-free conditions. The Guide for the Care and Use of Laboratory Animals (China) conducted all experimental procedures involving animals, and the Animal Research Ethics Committee of the Sir Run Run Shaw Hospital of Zhejiang University approved the protocols.

Female C57 mice were mated with male BALB/c mice to establish an allogeneic pregnancy model (Rowe et al., 2012). The day of the vaginal plug detection was considered day 0.5 of pregnancy. SBP was measured using a noninvasive mouse tailcuff BP analyzer (BP-2010A, Softron, Japan) at 12.5, and 16.5 days of gestation. Three random urine samples were collected after day 16.5 of gestation, and UACR was measured by urinary microalbumin (CH0101060, Maccura, China). Mice were euthanized on day 18.5 of gestation.

Reduction in uterine perfusion pressure mouse model

To construct a mouse model of PE, we ligated uterine arteries in pregnant mice on day 12.5 of gestation. Briefly, after 4% chloral hydrate was intraperitoneally injected for anesthesia, bilateral incisions were made on the back of the pregnant mice, and surgical sutures were used to reduce the blood flow of the bilateral uterine arcades.

Isolation of single cells from the mouse uterus and human placenta

Uterus from pregnant mice and placentas from volunteers were washed twice with ice-cold PBS and cut into small pieces. The tissues were digested with collagenase type IV (1 mg/ml, Sigma-Aldrich, USA) and DNase I (0.01 mg/ml, Sigma-Aldrich, U.S.A) in RPMI 1640 medium (Thermo Fisher Scientific) for 40 min at 200 rpm and 37 °C. The suspensions were strained through 70- μ m nylon mesh and centrifuged at 500 $\times$ g for 5 minutes. Leaving the supernatants, the cell pellets from human placentas need an extra purification by Ficoll (P4350, Solarbio, China) according to the manufacturer's instructions. Human CD4⁺ memory T cells were isolated using a Human Central and Effector Memory CD4⁺ T Cell Isolation Kit (17865, STEMCELL, Canada).

Table 5

Details of the individual with NP or PE included in the study

Parameters	NP (n=30)	PE (n=30)	p Value
Age (years)	31.13 ± 2.9747	31.50 ± 4.1693	0.0741
BMI (kg/m ²)	27.24 ± 2.8048	29.48±3.9295	0.0155
Gestational age (weeks)	38.68± 0.6038	35.53 ± 3.29	<0.0001
Number of living children	0.6000 ± 0.4900	0.4667 ± 0.6182	0.3665

Table 6

Details of the individual with early-onset preeclampsia or late-onset preeclampsia included in the study

Parameters	Early-onset preeclampsia (n=11)	Late-onset preeclampsia (n=19)	p Value
The gestational week when blood pressure begins to rise	28.82± 2.8220	36.05± 1.129	< 0.0001
Age (years)	32.45± 4.6768	32.36± 3.8472	0.9568
BMI (kg/m ²)	29.64± 2.6840	28.71±4.4506	0.5388
Gestational age (weeks)	33.45± 3.7887	36.77± 1.2171	0.0014
Number of living children	0.36± 0.5045	0.36 ± 0.4956	0.98

Adoptive transferred mouse model

Endogenous macrophages or CD4⁺ T cells were depleted by injecting PLX3397 (S7818, Selleck, USA), clodronate liposomes (40337ES08, YEASEN, China) or anti-CD4 antibody (BE0003-1, BioXcell, USA) intraperitoneally every three days starting from day 10.5 of gestation.

For the adoptive transferred mouse model, phycoerythrin-conjugated anti-mouse CD4 (12-0041-82, eBioscience, USA) and Pc5.5-conjugated anti-mouse CD44 (45-0441-82, eBioscience, USA) were used to label CD4⁺ memory-like T cells from uterus and placentas. PE-conjugated anti-mouse CD45 (E-AB-F1136D, Elabscience, China), FITC-conjugated anti-mouse F4/80 (11-4801-82, eBioscience, USA), and PE-Cyanine7-conjugated anti-mouse CD206 (141719, Biolegend, USA) were used to label anti-inflam Macs from uterus and placentas, and the remaining CD45⁺F480⁺CD206⁻ cells were considered pro-inflam Macs. The MoFlo Astrios EQ FACS Cell Sorter at Core Facilities, ZheJiang University School of Medicine (China), was used to obtain specific cells with a purity of > 96%. Briefly, $2-5 \times 10^5$ CD4⁺ CD44⁺ T cells, 5×10^5 pro-inflam Macs, or 5×10^5 anti-inflam Macs were injected to pregnant C57 mice intravenously at day 12.5 of gestation.

$2-5 \times 10^5$ CD45⁺CD4⁺ T cells from uterus and placentas of pregnant mice were sorted and transferred to pregnant C57 mice at day 11.5 of gestation after treating with 100nM BMS-754809 (HY10200; MedChemExpress; USA) or PBS for three days in vitro.

Mass cytometry by time of flight

Placental villi rather than chorionic plate and extraplacental membranes were used for CyToF in this study. 8×10^5 immune cells from NP (n=9), PE (n=8), GDM (n=8), GDM&PE (n=7) presented in **Table 1** [↗](#) were washed once with 1x PBS and then stained with 100μL of 2.5 μM cisplatin (Fluidigm) for 5min on ice to exclude dead cells, and then incubated in Fc receptor blocking solution before stained with surface antibodies cocktail for 30 min on ice. Cells were washed twice with FACS buffer (1xPBS+0.5%BSA) and fixed in 200μL of intercalation solution (Maxpar Fix and Perm Buffer containing 250nM 191/193Ir, Fluidigm) overnight. After fixation, cells were washed once with FACS buffer and then perm buffer (eBioscience), stained with intracellular antibodies cocktail for 30 min on ice. Cells were washed and resuspend with deionized water, adding into 20% EQ beads (Fluidigm), acquired on a mass cytometer (Helios, Fluidigm). Additionally, all PE samples enumerated in **Table 1** [↗](#) demonstrate a late-onset preeclampsia, with placental specimens being procured from patients more than 35 weeks of gestation and less than the 38 weeks of pregnancy.

As for data analysis, data of each sample were debarcoded from raw data using a doublet-filtering scheme with unique mass-tagged barcodes firstly. Each .fcs file generated from different batches were normalized through bead normalization method. Then we manually gate data using a FlowJo software to exclude to debris, dead cells and doublets, leaving live, single immune cells. Subsequently we applied the Phenograph clustering algorithm to all cells to partition the cells into distinct phenotypes based on marker expression levels. Cell type of each cluster according to its marker expression pattern on a heatmap of cluster vs marker was annotated. Finally, we used the dimensionality reduction algorithm t-SNE to visualize the high-dimensional data in two dimensions and show distribution of each cluster and marker expression and difference among each group or different sample type.

RNA sequencing

5×10^5 CD45⁺F4/80⁺CD206⁻ pro-inflam Macs (n=3) and 5×10^5 CD45⁺F4/80⁺CD206⁺ anti-inflam Macs (n=3) were isolated from the uterus and placentas of mice with RUPP and high throughput sequencing and bioinformatics analyses were conducted at Shenzhen Huada Gene Technology

Service Co. Ltd.(Shenzhen, China). Only differential genes with more than two-fold change and a corrected P value less than 0.05 were considered statistically significant.

Single-cell RNA sequencing (scRNA-seq)

Mice were injected with 5×10^5 CD45⁺F4/80⁺CD206⁻ pro-inflam Macs or 5×10^5 CD45⁺F4/80⁺CD206⁺ anti-inflam Macs at 12.5 days of gestation and euthanized on day 18.5 of gestation. CD45⁺ immune cells from the uterus and placenta were isolated from mice transferred pro-inflam Macs and anti-inflam Macs. scRNA-seq was conducted at PLTTECH Service Co. Ltd. (Hangzhou, China).

Firstly, the outcomes of cellular high-throughput sequencing were preserved in the FASTQ file format, facilitating a rigorous evaluation of the raw data. Subsequently, the data underwent debarcoding, enabling its alignment with the reference genome to precisely identify and sieve effective cells. This process culminated in the construction of a comprehensive gene-barcode expression matrix. For the statistical analysis and visualization of gene expression levels and Unique Molecular Identifier (UMI) counts, we leveraged the powerful Seurat software. Furthermore, we conducted a Clustering analysis rooted in the gene-barcode expression matrix, utilizing Seurat's capabilities in conjunction with Principal Component Analysis (PCA) for dimensionality reduction. This allowed us to scale down the clustering results and visually represent them in two dimensions using either Uniform Manifold Approximation and Projection (UMAP) or t-Distributed Stochastic Neighbor Embedding (tSNE). To gain deeper insights, we performed Marker genes analysis, which hinged on the clustering outcomes. This enabled us to pinpoint the genes that significantly distinguished each subgroup, thereby identifying their characteristic genes. Additionally, we analyzed and graphically represented the varying expression levels of these genes across different groups, providing a comprehensive understanding of their differences.

Isolation of NP-EVs and PE-EVs

NP-EVs and PE-EVs were isolated and identified using our published protocols (Jiang et al., 2021 [↗](#); X. Liu et al., 2022 [↗](#)). Briefly, after digesting the placental tissues from NP or PE, the suspensions were filtered through 100-µm nylon mesh and centrifuged at 3,000×g for 15 minutes. Then the supernatants were filtered with a 0.22-µm filter and centrifuged at 100,000×g for 1 h at 4 °C. Then the pellets of EVs were resuspended and centrifuged at 100,000×g once again. A BCA assay kit conducted Protein quantitation of EVs (23235, Thermo Fisher Scientific). To isolate the T-EVs, 500 µg of the EVs were incubated with 1 µg placental alkaline phosphatase antibody (SC-47691, Santa Cruz Biotechnology) at 4 °C overnight, then washed in the recommended buffer (PBS containing 2% exosome-free FBS and 1 mmol/L ethylenediaminetetraacetic acid), and centrifuged at 100,000×g for 1 hour at 4 °C. After resuspended, the total EVs were sorted by EasySep Mouse PE Positive Selection Kit II (17666, STEMCELL) to collect the final NP-EVs or PE-EVs.

Induction of CD4⁺ memory-like T cells

Macrophages were obtained after five days of culture of mononuclear cells isolated from human peripheral blood following previous protocols (X. Liu et al., 2022 [↗](#)). NP-EVs or PE-EVs in a 50 µg/mL concentration were added to macrophages and co-cultured at 37 °C for 8 h. The human CD4⁺ naïve T cells Isolation Kit (19555, STEMCELL, Canada) was used to isolate purified CD4⁺ naïve T cells from human peripheral blood cells according to the manufacturer's instructions. Then, CD4⁺ naïve T cells were co-cultured with EV-treated macrophages for six days. Flow cytometry was performed to measure the frequency of memory-like Th17 cells. For animal experiments, CD4⁺ naïve T cells sorted from the uterus and placentas of pregnant mice were cultured with the IGF1R inhibitor BMS-754807 at a concentration of 10 µM for three days before co-cultured with macrophages.

Flow cytometry

Single immune cells from the mice uterus were obtained following the method described above. After incubation with Cell Stimulation Cocktail (00-4975-93, Invitrogen, USA) for five hours at 37 °C, cells were collected and surface staining for either phycoerythrin-conjugated anti-mouse CD4 (12-0041-82, eBioscience, USA), APC-conjugated anti-mouse CD8 (100711, BioLegend, USA), phycoerythrin-Cy5.5-conjugated anti-mouse CD44 (45-0441-82, eBioscience, USA), APC-conjugated anti-mouse CD11b (101212, BioLegend, USA), FITC-conjugated anti-mouse Gr-1 (108406, BioLegend, USA), PE-Cyanine7-conjugated anti-mouse Ly6G (E-AB-F1108H, elabscience, China) or intracellular staining for 488-conjugated anti-mouse IL-17A (506910, BioLegend, USA) was performed according to the manufacturer's instructions.

Single placental and peripheral lymphocytes were obtained following the method described above. The Human Central and Effector Memory CD4⁺ T cell Isolation Kit (17865, STEMCELL, Canada) was used to obtain purified memory CD4⁺ T cells, according to the manufacturer's instructions. After incubation with Cell Stimulation Cocktail (00-4975-93, Invitrogen, USA) for five hours at 37 °C, cells were collected and surface staining for phycoerythrin-conjugated anti-human CD4 (2384240, eBioscience, USA), FITC-conjugated anti-human CD45RO (304204, BioLegend, USA), and phycoerythrin-Cy7-conjugated anti-human CCR7 (353227, BioLegend, USA) or intracellular staining for APC-conjugated anti-human IL-17A (17-7179-42, eBioscience, USA) and Foxp3 (320014, BioLegend, USA) was performed according to the manufacturer's instructions.

Immunofluorescence

Frozen sections of the placentas were permeabilized with PBS containing 0.5% Triton X-100 (PBST) for 20 min and incubated for 1 h with a blocking buffer in PBST. The sections were then incubated with anti-CD4 (sc-1176, Santa Cruz Biotechnology, China) and FITC-conjugated anti-human CD45RO (304204, BioLegend, USA) at 4 °C overnight, followed by Alexa Fluor 568-conjugated secondary antibodies (dilution: 1:200, Yeasen, China) for 1 h. The slides were counterstained with 4, 6-diamidino-2-phenylindole (DAPI, 1 µg/ml; Roche, Switzerland) for 20 min. Digital images were obtained using confocal fluorescence microscopy (ZEISS LSM 800, Germany). ImageJ software was used to quantify the fluorescence intensity from immunofluorescence (IF) images.

Statistical analysis

Data were analyzed using SPSS version 20. After the Shapiro-wilk test, the data were confirmed to be non-normal distribution. Kruskal-wallis test was used to compare the results of experiments with multiple groups. Student's t-test was used to compare the results of experiments with two groups. All data are presented as mean ± SEM (*P < 0.05, ** P < 0.01, *** P < 0.001; NS, not significant).

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Reviewer #1 (Public review):

Summary:

This study explores the immune microenvironment of the placenta in preeclampsia (PE), which is often accompanied by gestational diabetes mellitus (GDM). Using CyTOF, they found that placentas from PE cases showed increased frequencies of memory-like Th17 cells, memory-like CD8⁺ T cells, and pro-inflammatory macrophages, alongside decreased levels of anti-inflammatory macrophages and granulocyte myeloid-derived suppressor cells (gMDSCs) compared to normal pregnancies. Further analysis revealed a positive correlation between pro-inflammatory macrophages and the expanded T cell populations, and a negative

correlation with gMDSCs. Single-cell RNA sequencing provided mechanistic insights: transferring a specific subset of pro-inflammatory macrophages (F4/80⁺CD206⁻ with a distinct gene expression profile) from the uterus of PE mice to normal pregnant mice induced the formation of pathogenic memory-like Th17 cells via the IGF1-IGF1R pathway. This cellular interplay not only contributed to the development but also to the recurrence of PE. Additionally, these macrophages promoted the production of memory-like CD8⁺ T cells while inhibiting gMDSCs at the maternal-fetal interface, culminating in PE-like symptoms in mice. In conclusion, the study identifies a PE-specific immune cell network regulated by pro-inflammatory macrophages, offering new insights into the pathogenesis of preeclampsia.

Strengths:

Utilization of both human placental samples and multiple mouse models to explore the mechanisms linking inflammatory macrophages and T cells to preeclampsia (PE). Incorporation of cutting-edge and complementary techniques such as CyTOF, scRNA-seq, bulk RNA-seq, and flow cytometry.

Identification of specific immune cell populations and their roles in PE.
Demonstration of the adverse effects of pro-inflammatory macrophages and T cells on pregnancy outcomes through in vivo manipulations.

Comments on revised version:

Several weaknesses were addressed during revision by conducting additional experiments, clarifying the manuscript's text, and incorporating new data that was not initially included.

<https://doi.org/10.7554/eLife.100002.2.sa2>

Reviewer #2 (Public review):

Summary:

Fei, Lu, Shi, et al. present a thorough evaluation of the immune cell landscape in pre-eclamptic human placentas by single-cell multi-omics methodologies compared to normal control placentas. Based on their findings of elevated frequencies of inflammatory macrophages and memory-like Th17 cells, they employ adoptive cell transfer mouse models to interrogate the coordination and function of these cell types in pre-eclampsia immunopathology. They demonstrate the putative role of the IGF1-IGF1R axis as the key pathway by which inflammatory macrophages in the placenta skew CD4⁺ T cells towards an inflammatory IL-17A-secreting phenotype that may drive tissue damage, vascular dysfunction, and elevated blood pressure in pre-eclampsia, leaving researchers with potential translational opportunities to pursue this pathway in this indication.

They present a major advance to the field in their profiling of human placental immune cells from pre-eclampsia patients where most extant single-cell atlases focus on term versus preterm placenta, or largely examine trophoblast biology with a much rarer subset of immune cells. While the authors present vast amounts of data at both the protein and RNA transcript level, we, the reviewers, feel this manuscript is still in need of much more clarity in its main messaging, and more discretion in including only key data that supports this main message most effectively.

Strengths:

(1) This study combines human and mouse analyses and allows for some amount of mechanistic insight into the role of pro-inflammatory and anti-inflammatory macrophages in the pathogenesis of pre-eclampsia (PE), and their interaction with Th17 cells.

(2) Importantly, they do this using matched cohorts across normal pregnancy and common PE comorbidities like gestation diabetes (GDM).

(3) The authors have developed clear translational opportunities from these "big data" studies by moving to pursue potential IGF1-based interventions.

[Editors' note: the authors have provided responses to the previously identified weaknesses]

<https://doi.org/10.7554/eLife.100002.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1(Public review):

Strengths:

Utilization of both human placental samples and multiple mouse models to explore the mechanisms linking inflammatory macrophages and T cells to preeclampsia (PE). Incorporation of advanced techniques such as CyTOF, scRNA-seq, bulk RNA-seq, and flow cytometry.

Identification of specific immune cell populations and their roles in PE, including the IGF1-IGF1R ligand-receptor pair in macrophage-mediated Th17 cell differentiation. Demonstration of the adverse effects of pro-inflammatory macrophages and T cells on pregnancy outcomes through transfer experiments.

Weaknesses:

Comment 1. Inconsistent use of uterine and placental cells, which are distinct tissues with different macrophage populations, potentially confounding results.

Response1: We thank the reviewers' comments. We have done the green fluorescent protein (GFP) pregnant mice-related animal experiment, which was not shown in this manuscript. The wild-type (WT) female mice were mated with either transgenic male mice, genetically modified to express GFP, or with WT male mice, in order to generate either GFP-expressing pups (GFP-pups) or their genetically unmodified counterparts (WT-pups), respectively. Mice were euthanized on day 18.5 of gestation, and the uteri of the pregnant females and the placentas of the offspring were analyzed using flow cytometry. The majority of macrophages in the uterus and placenta are of maternal origin, which was defined by GFP negative. In contrast, fetal-derived macrophages, distinguished by their expression of GFP, represent a mere fraction of the total macrophage population. We have added the GFP pregnant mice-related data in uterine and placental cells (Line204-212).

Comment 2. Missing observational data for the initial experiment transferring RUPP-derived macrophages to normal pregnant mice.

Response 2: We thank the reviewers' comments. We have added the observational data (Figure 4-figure supplement 1D, 1E) and a corresponding description of the data (Line 198-203).

Comment 3. Unclear mechanisms of anti-macrophage compounds and their effects on placental/fetal macrophages.

Response 3: We thank the reviewers' comments. PLX3397, the inhibitor of CSF1R, which is needed for macrophage development (Nature. 2023, PMID: 36890231; Cell Mol Immunol. 2022, PMID: 36220994), we have stated that on Line 227-230. However, PLX3397 is a small molecule compound that possesses the potential to cross the placental barrier and affect fetal macrophages. We have discussed the impact of this factor on the experiment in the Discussion section (Line457-459).

Comment 4. Difficulty in distinguishing donor cells from recipient cells in murine single-cell data complicates interpretation.

Response 4: We thank the reviewers' comments. Upon analysis, we observed a notable elevation in the frequency of total macrophages within the CD45⁺ cell population. Then we subsequently performed macrophage clustering and uncovered a marked increase in the frequency of Cluster 0, implying a potential correlation between Cluster 0 and donor-derived cells. RNA sequencing revealed that the F480⁺CD206⁻ pro-inflammatory donor macrophages exhibited a Fcrl2⁺Ccl7⁺Ccl8⁺C1qa⁺C1qb⁺C1qc⁺ phenotype, which is consistent with the phenotype of cluster 0 in macrophages observed in single-cell RNA sequencing (Figure 4D and Figure 5E). Therefore, we believe that the donor cells should be cluster 0 in macrophages.

Comment 5. Limitation of using the LPS model in the final experiments, as it more closely resembles systemic inflammation seen in endotoxemia rather than the specific pathology of PE.

Response 5: We thank the reviewers' comments. Firstly, our other animal experiments in this manuscript used the Reduction in Uterine Perfusion Pressure (RUPP) mouse model to simulate the pathology of PE. However, the RUPP model requires ligation of the uterine arteries in pregnant mice on day 12.5 of gestation, which hinders T cells returning from the tail vein from reaching the maternal-fetal interface. In addition, this experiment aims to prove that CD4⁺ T cells are differentiated into memory-like Th17 cells through IGF-1R receptor signaling to affect pregnancy by clearing CD4⁺ T cells in vivo with an anti-CD4 antibody followed by injecting IGF-1R inhibitor-treated CD4⁺ T cells. And we proved that injection of RUPP-derived memory-like CD4⁺ T cells into pregnant mice induces PE-like symptoms (Figure 6F-6H). In summary, the application of the LPS model in the final experiments does not affect the conclusions.

Reviewer #2 (Public review):

Strengths:

(1) This study combines human and mouse analyses and allows for some amount of mechanistic insight into the role of pro-inflammatory and anti-inflammatory macrophages in the pathogenesis of pre-eclampsia (PE), and their interaction with Th17 cells.

(2) Importantly, they do this using matched cohorts across normal pregnancy and common PE comorbidities like gestation diabetes (GDM).

(3) The authors have developed clear translational opportunities from these "big data" studies by moving to pursue potential IGF1-based interventions.

Weaknesses:

(1) Clearly the authors generated vast amounts of multi-omic data using CyTOF and single-cell RNA-seq (scRNA-seq), but their central message becomes muddled very quickly. The reader has to do a lot of work to follow the authors' multiple lines of inquiry rather than smoothly following along with their unified rationale. The title description tells fairly little about the substance of the study. The manuscript is very challenging to follow. The paper would benefit from substantial reorganizations and editing for grammatical and spelling errors. For example, RUPP is introduced in Figure 4 but in the text not defined or even talked about what it is until Figure 6. (The figure comparing pro- and anti-inflammatory macrophages does not add much to the manuscript as this is an expected finding).

Response 1: We thank the reviewers' comments. According to the reviewer's suggestion, we have made the necessary revisions. Firstly, the title of the article has been modified to be more specific. We also introduce the RUPP mouse model when interpreted Figure 4-figure supplement 1. Thirdly, We have moved the images of Figure 7 to the Figure 6-figure supplement 2 make them easier to follow. Finally, we diligently corrected the grammatical and spelling errors in the article. As for the figure comparing pro- and anti-inflammatory macrophages, the Editor requested a more comprehensive description of the macrophage phenotype during the initial submission. As a result, we conducted the transcriptome RNA-seq of both uterine-derived pro-inflammatory and anti-inflammatory macrophages and conducted a detailed analysis of macrophages in scRNA-seq.

Comment 2. The methods lack critical detail about how human placenta samples were processed. The maternal-fetal interface is a highly heterogeneous tissue environment and care must be taken to ensure proper focus on maternal or fetal cells of origin. Lacking this detail in the present manuscript, there are many unanswered questions about the nature of the immune cells analyzed. It is impossible to figure out which part of the placental unit is analyzed for the human or mouse data. Is this the decidua, the placental villi, or the fetal membranes? This is of key importance to the central findings of the manuscript as the immune makeup of these compartments is very different. Or is this analyzed as the entirety of the placenta, which would be a mix of these compartments and significantly less exciting?

Response 2: We thank the reviewers' comments. Placental villi rather than fetal membranes and decidua were used for CyTOF in this study. This detail about how human placenta samples were processed have been added to the Materials and Methods section (Line564-576).

Comment 3. Similarly, methods lack any detail about the analysis of the CyTOF and scRNAseq data, much more detail needs to be added here. How were these clustered, what was the QC for scRNAseq data, etc? The two small paragraphs lack any detail.

Response 3: We thank the reviewers' comments. The details about the analysis of the CyTOF (Line577-586) and scRNAseq (Line600-615) data have been added in the Materials and Methods section.

Comment 4. There is also insufficient detail presented about the quantities or proportions of various cell populations. For example, gdT cells represent very small proportions of the CyTOF plots shown in Figures 1B, 1C, & 1E, yet in Figures 2I, 2K, & 2K there are many gdT cells shown in subcluster analysis without a description of how many cells are actually represented, and where they came from. How were biological replicates normalized for fair statistical comparison between groups?

Response 4: We thank the reviewers' comments. In our study, approximately 8×10^5 cells were collected per group for analysis using CyTOF. Of these, about 10% (8×10^4 cells per group) were utilized to generate Figure 1B. As depicted in Figure 1B, gdT cells constitute roughly 1% of each group, with specific percentages as follows: NP group (1.23%), PE group (0.97%), GDM group (0.94%), and GDM&PE group (1.26%), which equates to approximately 800 cells per group. For the subsequent gdT cell analysis presented in Figure 2I, we employed data from all cells within each group to construct the tSNE maps, comprising approximately 8000 cells per group. Consequently, it may initially appear that the number of gdT cells is significantly higher than what is shown in Figure 1B. To clarify this, we have included pertinent explanations in the figure legend. Given the relatively low proportions of gdT cells, we did not pursue further investigations of these cells in subsequent experiments. Following your suggestion, we have relocated this result to the supplementary materials, where it is now presented as Figure 2-figure supplement 1D-E.

The number of biological replicates (samples) is consistent with Figure 1, and this information has been added to the figure legend.

Comment 5. The figures themselves are very tricky to follow. The clusters are numbered rather than identified by what the authors think they are, the numbers are so small, that they are challenging to read. The paper would be significantly improved if the clusters were clearly labeled and identified. All the heatmaps and the abundance of clusters should be in separate supplementary figures.

Response 5: We thank the reviewers' comments. Based on your suggestions, we have labeled and defined the Clusters (Figure 2A, 2F, Figure 3A, Figure 5C and Figure 6A). Additionally, we have moved most of the heatmaps to the supplementary materials.

Comment 6. The authors should take additional care when constructing figures that their biological replicates (and all replicates) are accurately represented. Figure 2H-2K shows N=10 data points for the normal pregnant (NP) samples when clearly their Table 1 and test denote they only studied N=9 normal subjects.

Response 6: We thank the reviewers' careful checking. During our verification, we found that one sample in the NP group had pregnancy complications other than PE and GDM. The data in Figure 2H-2K was not updated in a timely manner. We have promptly updated this data and reanalyze it.

Comment 7. There is little to no evaluation of regulatory T cells (Tregs) which are well known to undergird maternal tolerance of the fetus, and which are well known to have overlapping developmental trajectory with RORgt+ Th17 cells. We recommend the authors evaluate whether the loss of Treg function, quantity, or quality leaves CD4+ effector T cells more unrestrained in their effect on PE phenotypes. References should include, accordingly: PMID: PMC6448013 / DOI: 10.3389/fimmu.2019.00478; PMC4700932 / DOI: 10.1126/science.aaa9420.

Response 7: We thank the reviewers' comments. We have done the Treg-related animal experiment, which was not shown in this manuscript. We have added the Treg-related data in Figure 6F-6H. The injection of CD4⁺CD44⁺ T cells derived from RUPP mouse, characterized by a reduced frequency of Tregs, could induce PE-like symptoms in pregnant mice (Line297-304). Additionally, we have added a necessary discussion about Tregs and cited the literature you mentioned (Line433-439).

Comment 8. In discussing gMDSCs in Figure 3, the authors have missed key opportunities to evaluate bona fide Neutrophils. We recommend they conduct FACS or CyTOF staining

including CD66b if they have additional tissues or cells available. Please refer to this helpful review article that highlights key points of distinguishing human MDSC from neutrophils: <https://doi.org/10.1038/s41577-024-01062-0>. This will both help the evaluation of potentially regulatory myeloid cells that may suppress effector T cells as well as aid in understanding at the end of the study if IL-17 produced by CD4⁺ Th17 cells might recruit neutrophils to the placenta and cause ROS immunopathology and fetal resorption.

Response 8: We thank the reviewers' comments. Although we do not have additional tissues or cells available to conduct FACS or CyTOF staining, including for CD66b, we have utilized CD15 and CD66b antibodies for immunofluorescence stain of placental tissue, and our findings revealed a pronounced increase in the proportion of neutrophils among PE patients, fostering the hypothesis that IL-17A produced by Th17 cells might orchestrate the migration of neutrophils towards the placental milieu (Figure 6-figure supplement 2F; Line 325-328). We have cited these references and discussed them in the Discussion section (Line 459-465).

Comment 9. Depletion of macrophages using several different methodologies (PLX3397, or clodronate liposomes) should be accompanied by supplementary data showing the efficiency of depletion, especially within tissue compartments of interest (uterine horns, placenta). The clodronate piece is not at all discussed in the main text. Both should be addressed in much more detail.

Response 9: We thank the reviewers' comments. We already have the additional data on the efficiency of macrophage depletion involving PLX3397 and clodronate liposomes, which were not present in this manuscript, and we'll add it to the Figure 4-figure supplement 2A,2B. The clodronate piece is mentioned in the main text (Line236-239), but only briefly described, because the results using clodronate we obtained were similar to those using PLX3397.

Comment 10. There are many heatmaps and tSNE / UMAP plots with unhelpful labels and no statistical tests applied. Many of these plots (e.g. Figure 7) could be moved to supplemental figures or pared down and combined with existing main figures to help the authors streamline and unify their message.

Response 10: We thank the reviewers' comments. We have moved the images of Figure 7 to the Figure 6-figure supplement 2. We also have moved most of the heatmaps to the supplementary materials.

Comment 11. There are claims that this study fills a gap that "only one report has provided an overall analysis of immune cells in the human placental villi in the presence and absence of spontaneous labor at term by scRNA-seq (Miller 2022)" (lines 362-364), yet this study itself does not exhaustively study all immune cell subsets...that's a monumental task, even with the two multi-omic methods used in this paper. There are several other datasets that have performed similar analyses and should be referenced.

Response 11: We thank the reviewers' comments. We have search for more literature and reference additional studies that have conducted similar analyses (Line382-393).

Comment 12. Inappropriate statistical tests are used in many of the analyses. Figures 1-2 use the Shapiro-Wilk test, which is a test of "goodness of fit", to compare unpaired groups. A Kruskal-Wallis or other nonparametric t-test is much more appropriate. In other instances, there is no mention of statistical tests (Figures 6-7) at all. Appropriate tests should be added throughout.

Response 12: We thank the reviewers' comments. As stated in the Statistical Analysis section (lines 672-676), the Kruskal-Wallis test was used to compare the results of experiments with multiple groups. Comparisons between the two groups in Figures 5 were conducted using Student's t-test. The aforementioned statistical methods have been included in the figure legends.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Overall, the study has several strengths, including the use of human samples and animal models, as well as the incorporation of multiple cutting-edge techniques. However, there are some significant issues with the murine model experiments that need to be addressed:

Comment 1. The authors are not consistent in their use of or focus on uterine and placental cells. These are distinct tissues, and numerous prior reports have indicated differences in the macrophage populations of these tissues, due in part to the predominantly maternal origin of macrophages in the uterus and the largely fetal origin of those in the placenta. The rationale for switching between uterine and placental cells in different experiments is not clear, and the inclusion of cells from both (such as in the bulk RNAseq experiments) could be potentially confounding.

Response 1: We thank the reviewers' comments. We have done the green fluorescent protein (GFP) pregnant mice-related animal experiment, which was not shown in this manuscript. The wild-type (WT) female mice were mated with either transgenic male mice, genetically modified to express GFP, or with WT male mice, in order to generate either GFP-expressing pups (GFP-pups) or their genetically unmodified counterparts (WT-pups), respectively. Mice were euthanized on day 18.5 of gestation, and the uteri of the pregnant females and the placentas of the offspring were analyzed using flow cytometry. The majority of macrophages in the uterus and placenta are of maternal origin, which was defined by GFP negative. In contrast, fetal-derived macrophages, distinguished by their expression of GFP, represent a mere fraction of the total macrophage population, signifying their inconsequential or restricted presence amidst the broader cellular landscape. We have added the GFP pregnant mice-related data in Figure 4-figure supplement 1D-1E to explain the different macrophage populations in the uterine and placental cells.

Comment 2. The observational data for the initial experiment transferring RUPP-derived macrophages to normal pregnant mice (without any other manipulations) seems to be missing. They do not seem to be presented in Figure 4 where they are expected based on the results text.

Response 2: We thank the reviewers' comments. We thank the reviewers' comments. We have added the observational data (Figure 4-figure supplement 1D, 1E) and a corresponding description of the data (Line 198-203).

Comment 3. The action of the anti-macrophage compounds is not well explained, nor are their mechanisms validated as affecting or not affecting the placental/fetal macrophage populations. It is important to clarify whether the macrophages are depleted or merely inhibited by these treatments, and it is absolutely critical to determine whether these treatments are affecting placental/fetal macrophage populations (the latter indicative of placental transfer), given the focus on placental macrophages.

Response 3: We thank the reviewers' comments. PLX3397, the inhibitor of CSF1R, which is needed for macrophage development (Nature. 2023, PMID: 36890231; Cell Mol Immunol. 2022, PMID: 36220994), we have stated that on Line227-230. However, PLX3397 is a small molecule compound that possesses the potential to cross the placental barrier and affect fetal macrophages. We will discuss the impact of this factor on the experiment in the Discussion section (Line457-459).

Comment 4. The interpretation of the murine single-cell data is hampered by the lack of means for distinguishing donor cells from recipient cells, which is important when seeking to identify the influence of the donor cells.

Response 4: We thank the reviewers' comments. Upon analysis, we observed a notable elevation in the frequency of total macrophages within the CD45⁺ cell population. Then we subsequently performed macrophage clustering and uncovered a marked increase in the frequency of Cluster 0, implying a potential correlation between Cluster 0 and donor-derived cells. RNA sequencing revealed that the F480⁺CD206⁻ pro-inflammatory donor macrophages exhibited a FcγR2⁺Ccl7⁺Ccl8⁺C1qa⁺C1qb⁺C1qc⁺ phenotype, which is consistent with the phenotype of cluster 0 in macrophages observed in single-cell RNA sequencing (Figure 4D and Figure 5E). Therefore, the donor cells should be in cluster 0 in macrophages.

Comment 5. The switch to the LPS model in the final experiments is a limitation, as this model more closely resembles the systemic inflammation seen in endotoxemia rather than the specific pathology of preeclampsia (PE). While this is not an exhaustive list, the number of weaknesses in the experimental design makes it difficult to evaluate the findings comprehensively.

Response 5: We thank the reviewers' comments. Firstly, our other animal experiments in this manuscript used the RUPP mouse model to simulate the pathology of PE. However, the RUPP model requires ligation of the uterine arteries in pregnant mice on day 12.5 of gestation, which hinders T cells returning from the tail vein from reaching the maternal-fetal interface. In addition, this experiment aims to prove that CD4⁺ T cells are differentiated into memory-like Th17 cells through IGF-1R receptor signaling to affect pregnancy by clearing CD4⁺ T cells in vivo with an anti-CD4 antibody followed by injecting IGF-1R inhibitor-treated CD4⁺ T cells. We proved that injection of RUPP-derived memory-like CD4⁺ T cells into pregnant rats induces PE-like symptoms (Figure 6F-6H). In summary, applying the LPS model in the final experiments does not affect the conclusions.

Minor comments:

Comment 1. Introduction, Lines 67-74: The phrasing here is unclear as to the roles that each mentioned immune cell subset is playing in preeclampsia. Given the statement "Elevated levels of maternal inflammation...", does this imply that the numbers of all mentioned immune cell subsets are increased in the maternal circulation? If not, please consider rewording this.

Response 1: We thank the reviewers' comments. We have revised the manuscript as follows: Currently, the pivotal mechanism underpinning the pathogenesis of preeclampsia is widely acknowledged to involve an increased frequency of pro-inflammatory M1-like maternal macrophages, along with an elevation in Granulocytes capable of superoxide generation, CD56⁺ CD94⁺ natural killer (NK) cells, CD19⁺CD5⁺ B1 lymphocytes, and activated γδ T cells. Conversely, this pathological process is accompanied by a notable decrease in the frequency of anti-inflammatory M2-like macrophages and NKp46⁺ NK cells (Line67-77).

Comment 2. Introduction, Lines 67-80: Is the involvement of the described immune cell subsets largely ubiquitous to preeclampsia? Recent multi-omic studies suggest that preeclampsia is a heterogeneous condition with different subsets, some more biased towards systemic immune activation than others. Thus, it is important to clarify whether the involvement of specific immune subsets is generally observed or more specific.

Response 2: We thank the reviewers' comments. We have added a new paragraph as follows: Moreover, as PE can be subdivided into early- and late-onset PE diagnosed before 34 weeks or from 34 weeks of gestation, respectively. Research has revealed that among the myriad of cellular alterations in PE, pro-inflammatory M1-like macrophages and intrauterine B1 cells display an augmented presence at the maternal-fetal interface of both early-onset and late-onset PE patients. Decidual natural killer (dNK) cells and neutrophils emerge as paramount contributors, playing a more crucial role in the pathogenesis of early-onset PE than late-onset PE (Front Immunol. 2020. PMID: 33013837) (Line83-89).

Comment 3. Introduction, Lines 81-86: The point of this short paragraph is not clear; the authors mention two very specific cellular interactions without explaining why.

Response 3: In the previous paragraph, we uncovered a heightened inflammatory response among multiple immune cells in patients with PE, yet the intricate interplay between these individual immune cells has been seldom elucidated in the context of PE patient. This is precisely why we delve into the realm of specific immune cellular interactions in relation to other pregnancy complications in this paragraph (Line91-98).

Comment 4. Methods: What placental tissues (e.g., villous tree, chorionic plate, extraplacental membranes) were included for CyTOF analysis? Was any decidual tissue (e.g., basal plate) included? Please clarify.

Response 4: Placental villi rather than chorionic plate and extraplacental membranes were used for CyTOF in this study. The relevant content has been incorporated into the "Materials and Methods" section (Line564-576).

Comment 5. Results, Table 1: The authors should clarify that all PE samples were not full term (i.e., were less than 37 weeks of gestation), which is to be expected. In addition, were the PE cases all late-onset PE?

Response 5: All PE samples enumerated in Table 1 demonstrate a late-onset preeclampsia, with placental specimens being procured from patients more than 35 weeks of gestation and less than the 38 weeks of pregnancy. The relevant content has been incorporated into the "Materials and Methods" section (Line574-576).

Comment 6. Results, Figure 1: Are the authors considering the identified Macrophage cluster as being largely fetal (e.g., Hofbauer cells)? This also depends on whether any decidual tissue was included in the placental samples for CyTOF.

Response 6: Firstly, the specimens subjected to CyTOF analysis were devoid of decidual tissue and exclusively comprised placental villi. Secondly, the Macrophage cluster in Figure 1 undeniably encompasses Hofbauer cells, and we considering fetal-derived macrophages likely constituting the substantial proportion of the cellular population. However, a limitation of the CyTOF technique lies in its inability to discern between maternal and fetal origins of these cells, thereby precluding a definitive distinction.

Comment 7. Results, Figure 2C: Did the authors validate other T-cell subset markers (e.g., Th1, Th2, Th9, etc.)?

Response 7: In this study, we did not validate additional T-cell subset markers presented in Figure 2C, recognizing the potential for deeper insights. As we embark on our subsequent research endeavors, we aim to meticulously explore and characterize the intricate changes in diverse T-cell populations at the maternal-fetal interface, with a particular focus on preeclampsia patients, thereby advancing our understanding of this complex condition.

Comment 8. Results, Figure 2D: Where were the detected memory-like T cells located in the placenta? Did they cluster in certain areas or were they widely distributed?

Response 8: Upon a thorough re-evaluation of the immunofluorescence images specific to the placenta, we observed a notable preponderance of memory-like T cells residing within the placental sinusoids (Line135-139).

Comment 9. Results, Figure 2E: I would suggest separating the two plots so that the Y-axis can be expanded for TIM3, as it is impossible to view the medians currently.

Response 9: We thank the reviewers' comments. We have made the adjustment to Figure 2E according to the reviewers' suggestions.

Comment 10. Results, Lines 138-140: Do the authors consider that the altered T-cells are largely resident cells of the placenta or newly invading/recruited cells? The clarification of distribution within the placental tissues as mentioned above would help answer this.

Response 10: Our analysis revealed the presence of memory-like T cells within the placental sinusoids, as evident from the immunofluorescence examination of placental tissues. Consequently, these T cells may represent recently recruited cellular entities, traversing the placental vasculature and integrating into this unique maternal-fetal microenvironment (Line135-139).

Comment 11. Results, Figure 3C: Has a reduction of gMDSCs (or MDSCs in general) been previously reported in PE?

Response 11: Myeloid-derived suppressor cells (MDSCs) constitute a diverse population of myeloid-derived cells that exhibit immunosuppressive functions under various conditions. Previous reports have documented a decrease in the levels of gMDSCs from peripheral blood or umbilical cord blood among patients with preeclampsia (Am J Reprod Immunol. 2020, PMID: 32418253; J Reprod Immunol. 2018, PMID: 29763854; Biol Reprod. 2023, PMID: 36504233). Nevertheless, there was no documented reports thus far on the alterations and specific characteristics in gMDSCs within the placenta of PE patients.

Comment 12. Results, Figure 3D-E: It is not clear what new information is added by the correlations, as the increase of both cluster 23 in CD11b+ cells and cluster 8 in CD4+ T cells in PE cases was already apparent. Are these simply to confirm what was shown from the quantification data?

Response 12: Despite the evident increase in both cluster 23 within CD11b⁺ cells and cluster 8 within CD4⁺ T cells in PE cases, the existence of a potential correlation between these two clusters remains elusive. To gain insight into this question, we conducted a Pearson correlation analysis, which is presented in Figure 3D-E, revealing a positive correlation between the two clusters.

Comment 13. Results, Figure 4A: Please clarify in the results text that the RNA-seq of macrophages from RUPP mice was performed prior to their injection into normal pregnant mice.

Response 13: We thank the reviewers' comments. We have updated Figure 4A according to the reviewers' suggestions.

Comment 14. Results / Methods, Figure 4: For the transfer of macrophages from RUPP mice into normal mice, why were the uterine tissues included to isolate cells? The uterine macrophages will be almost completely maternal, as opposed to the largely fetal placental macrophages, and despite the sorting for specific markers these are likely distinct subsets that have been combined for injection. This could potentially impact the differential gene expression analysis and should be accounted for. In addition, did murine placental samples include decidua? This should be clarified.

Response 14: We thank the reviewers' comments. For our experimental design involving human samples, we meticulously selected placental tissue as the primary focus. Initially, we aimed for uniformity by contemplating the utilization of mouse placenta. However, a pivotal revelation emerged from the GFP pregnant mice-related data in Figure 4-figure supplement 1D,1E: the uterus and placenta of mice are predominantly populated by maternal macrophages, with fetal macrophages virtually absent, marking a notable divergence from the human scenario. Furthermore, the uterine milieu exhibits a macrophage concentration exceeding 20% of total cellular composition, whereas in the placenta, this proportion dwindles to less than 5%, underscoring a distinct distribution pattern. Given these discrepancies and considerations, we incorporated mouse uterine tissues into our protocol to isolate cells, ensuring a more comprehensive and informative exploration that acknowledges the inherent differences between human and mouse placental biology.

Comment 15. Results, Lines 186-187: I think the figure citation should be Figure 4D here.

Response 15: We thank the reviewers' careful checking. We have revised and updated Figure 4 accordingly.

Comment 16. Results, Figure 4: Where are the results of the injection of anti-inflammatory and pro-inflammatory macrophages into normal mice? This experiment is mentioned in Figure 4A, but the only results shown in Figure 4 are with the PLX3397 depletion.

Response 16: The aim of this experiment in figure 4 is to conclusively ascertain the influence of pro-inflammatory and anti-inflammatory macrophages on the other immune cells within the maternal-fetal interface, as well as their implications for pregnancy outcomes. To achieve this, we employed a strategic approach involving the administration of PLX3397, a compound capable of eliminating the preexisting macrophages in mice. Subsequently, anti-inflam or pro-inflam macrophages were injected to these mice, thereby eliminating the confounding influence of the native macrophage population. This methodology allows for a more discernible observation of the specific effects these two types of macrophages exert on the immune landscape at the maternal-fetal interface and their ultimate impact on pregnancy outcomes.

Comment 17. Results, Lines 189-190: Does PLX3397 inhibit macrophage development/signaling/etc. or result in macrophage depletion? This is an important distinction. If depletion is induced, does this affect placental/fetal macrophages or just maternal macrophages?

Response 17: We thank the reviewers' comments. We have updated the additional data on the efficiency of macrophage depletion involving PLX3397 in Figure 4-figure supplement 2A. PLX3397 is a small molecule compound that possesses the potential to cross the placental barrier and affect fetal macrophages. We have discussed the impact of this factor on the experiment in the Discussion section (Line457-459).

Comment 18. Results, Lines 197-198: Similarly, does clodronate liposome administration affect only maternal macrophages, or also placental/fetal macrophages?

Response 18: We thank the reviewers' comments. We have updated the additional data on the efficiency of macrophage depletion involving Clodronate Liposomes in Figure 4-figure supplement 2B. Clodronate Liposomes, which are intricate vesicles encapsulating diverse substances, while only small molecule compounds possess the potential to cross the placental barrier. Consequently, we hold the view that the influence of these liposomes is likely confined to the maternal macrophages (Artif Cells Nanomed Biotechnol. 2023. PMID: 37594208).

Comment 19. Results, Line 206: A minor point, but consider continuing to refer to the preeclampsia model mice as RUPP mice rather than PE mice.

Response 19: We thank the reviewers' comments. We have revised and updated this section accordingly.

Comment 20. Results / Methods, Figure 5: For these experiments, why did the authors focus on the mouse uterus?

Response 20: We have previously addressed this query in our Response 14. We incorporated mouse uterine tissues for cell isolation due to the profound differences in placental biology between humans and mice.

Comment 21. Results, Figure 5: Did the authors have a means of distinguishing the transferred donor cells from the recipient cells for their single-cell analysis? If the goal is to separate the effects of the macrophage transfer on other uterine immune cells, then it would be important to identify and separate the donor cells.

Response 21: We thank the reviewers' comments. Upon analysis, we observed a notable elevation in the frequency of total macrophages within the CD45⁺ cell population. Then we subsequently performed macrophage clustering and uncovered a marked increase in the frequency of Cluster 0, implying a potential correlation between Cluster 0 and donor-derived cells. RNA sequencing revealed that the F480⁺CD206⁻ pro-inflammatory donor macrophages exhibited a Fcrl2⁺Ccl7⁺Ccl8⁺C1qa⁺C1qb⁺C1qc⁺ phenotype, which is consistent with the phenotype of cluster 0 in macrophages observed in single-cell RNA sequencing (Figure 4D and Figure 5E). Therefore, the donor cells should be in cluster 0 in macrophages.

Comment 22. Results, Lines 247-248: While the authors have prudently noted that the observed T-cell phenotypes are merely suggestive of immunosuppression, any claims regarding changes in the immunosuppressive function after macrophage transfer would require functional studies of the T cells.

Response 22: We thank the reviewers' comments. Upon revisiting and meticulously reviewing the pertinent literature, we have refined our terminology, transitioning from 'immunosuppression' to 'immunomodulation', thereby enhancing the accuracy and precision of our Results (Line285-287).

Comment 23. Results, Figure 6G: The observation of worsened outcomes and PE-like symptoms after T-cell transfer is interesting, but other models of PE induced by the administration of Th1-like cells have already been reported. Are the authors' findings consistent with these reports? These findings are strengthened by the evaluation of second-pregnancy outcomes following the transfer of T cells in the first pregnancy.

Response 23: We thank the reviewers' comments. As we verified in Figure 6F-6H, the injection of CD4⁺CD44⁺ T cells derived from RUPP mouse, characterized by a reduced frequency of Tregs and an increased frequency of Th17 cells, could induce PE-like symptoms in pregnant mice. In line with other studies, which have implicated Th1-like cells in the manifestation of PE-like symptoms, we posit a novel hypothesis: beyond Th1 cells, Th17 cells also have the potential to induce PE-like symptoms.

Comment 24. Results, Lines 327-337: The disease model implied by the authors here is not clear. Given that the authors' human findings are in the placental macrophages, are the authors proposing that placental macrophages are induced to an M1 phenotype by placenta-derived EVs? Please elaborate on and clarify the proposed model.

Response 24 In the article authored by our team, titled "Trophoblast-Derived Extracellular Vesicles Promote Preeclampsia by Regulating Macrophage Polarization" published in Hypertension (Hypertension. 2022, PMID: 35993233), we employed trophoblast-derived extracellular vesicles isolated from PE patients as a means to induce an M1-like macrophage phenotype in macrophages from human peripheral blood in vitro. Consequently, in the present study, we have directly leveraged this established methodology to induce pro-inflammatory macrophages.

Comment 25. Results / Methods, Figure 8E-H: What is the reasoning for switching to an LPS model in this experiment? LPS is less specific to PE than the RUPP model.

Response 25: We thank the reviewers' comments. Firstly, our other animal experiments in this manuscript used the RUPP mouse model to simulate the pathology of PE. However, the RUPP model requires ligation of the uterine arteries in pregnant mice on day 12.5 of gestation, which hinders T cells returning from the tail vein from reaching the maternal-fetal interface. In addition, this experiment aims to prove that CD4⁺ T cells are differentiated into memory-like Th17 cells through IGF-1R receptor signaling to affect pregnancy by clearing CD4⁺ T cells in vivo with an anti-CD4 antibody followed by injecting IGF-1R inhibitor-treated CD4⁺ T cells. And we proved that injection of RUPP-derived memory-like CD4⁺ T cells into pregnant mice induces PE-like symptoms (Figure 6). In summary, the application of the LPS model in the final experiments does not affect the conclusions.

Comment 26. Discussion: What do the authors consider to be the origins of the inflammatory cells associated with PE onset? Are these maternal cells invading the placental tissues, or are these placental resident (likely fetal) cells?

Response 26: We thank the reviewers' comments. Numerous reports have consistently observed the presence of inflammatory cells and factors in the maternal peripheral blood and placenta tissues of PE patients, fostering the prevailing notion that the progression of PE is intricately linked to the maternal immune system's inflammatory response towards the fetus. Nevertheless, intriguing findings from single-cell RNA sequencing, analyzed through bioinformatic methods, have challenged this perspective (Elife. 2019. PMID: 31829938 ; Proc Natl Acad Sci U S A. 2017.PMID: 28830992). These studies reveal that the placenta harbors not just immune cells of maternal origin but also those of fetal origin, raising questions about

whether these are maternal cells infiltrating placental tissues or resident (possibly fetal) placental cells. Further investigation is imperative to elucidate this complex interplay.

Comment 27. Discussion: Given the observed lack of changes in the GDM or GDM+PE groups, do the authors consider that GDM represents a distinct pathology that can lead to secondary PE, and thus is different from primary PE without GDM?

Response 27: It's possible. Though previous studies reported GDM is associated with aberrant maternal immune cell adaption the findings remained controversial. It seems that GDM does not induce significant alterations in placental immune cell profile in our study, which made us pay more attention to the immune mechanism in PE. However, it is confusing for the reasons why individuals with GDM&PE were protected from the immune alterations at the maternal fetal interface. Limited placental samples in the GDM&PE group can partly explain it, for it is hard to collect clean samples excluding confounding factors. A study reported that macrophages in human placenta maintained anti-inflammatory properties despite GDM (Front Immunol, 2017, PMID: 28824621). Barke et al. also found that more CD163⁺ cells were observed in GDM placentas compared to normal controls (PLoS One, 2014, PMID: 24983948). Thus, GDM is likely to have a protective property in the placental immune environment when the individuals are complicated with PE.

Reviewer #2 (Recommendations for the authors):

Comment 1. IF images need to be quantified.

Response 1: We thank the reviewers' comments. We have quantified and calculated the fluorescence intensity and added it in Figure 2D.

Comment 2. Cluster 12 in Figure 3 is labeled as granulocytes but listed under macrophages.

Response 2: We thank the reviewers' careful checking. We have revised and updated Figure 3A.

Comment 3. Figure 4 labels in the text and figure do not match, no 4G in the figure.

Response 3: We thank the reviewers' careful checking. The figure labels of Figure 4 have been revised and updated.

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